Networking 1 · Complete Instructor Guide

Comprehensive Lecture & Discussion Handbook for All 13 Curriculum Topics

This master handbook compiles step-by-step teaching notes, everyday analogies, talking points, and classroom check-for-understanding questions for every lesson slide. Slide 1 (Title), Slide 2 (Objectives), and the Summary slide are excluded from each topic.

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TOPIC 1 OF 13

Topic 1: Introduction to Networking and Network Architecture

Discover everyday networking, classify network coverage and architectures, and follow a simple exchange between devices.

Curriculum: Networking 1

Excluded: Title (Slide 1), Objectives (Slide 2), and Summary (Final Slide)

Discussion Units: 19 Slides

Slide 3 of 22

What Is a Computer Network?

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A computer network is a system of two or more connected devices that exchange data and share resources. Wired or wireless links carry the data using agreed communication rules called protocols.

- Connected devices: computers, phones, servers, and printers can all participate in a network; the devices do not have to be identical.
- Communication: a device sends data through a connection, and another device receives and interprets it using compatible communication rules.
- Shared access: connected users can exchange messages, use shared files, or send work to a network printer. A local network can provide these services without Internet access.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Discover everyday networking, classify network coverage and architectures, and follow a simple exchange between devices.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Connected devices: computers, phones, servers, and printers can all participate in a network; the devices do not have to be identical.
- Communication: a device sends data through a connection, and another device receives and interprets it using compatible communication rules.
- Shared access: connected users can exchange messages, use shared files, or send work to a network printer.
- A local network can provide these services without Internet access.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of what is a computer network?.

Key Takeaway: A computer network is a system of two or more connected devices that exchange data and share resources. Wired or wireless links carry the data using agreed communication rules called protocols.

Slide 4 of 22

Everyday Uses of Networks

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Networks let people communicate and use services hosted on other devices. Messaging, online learning, shared files, and shared printers are familiar examples.

- Messaging: a phone sends a message to another person through a connected service.
- Online learning: a student accesses course materials hosted by a school or Internet service.
- Shared resources: several computers use the same printer or store files on a shared host.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Discover everyday networking, classify network coverage and architectures, and follow a simple exchange between devices.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Messaging: a phone sends a message to another person through a connected service.
- Online learning: a student accesses course materials hosted by a school or Internet service.
- Shared resources: several computers use the same printer or store files on a shared host.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of everyday uses of networks.

Key Takeaway: Networks let people communicate and use services hosted on other devices. Messaging, online learning, shared files, and shared printers are familiar examples.

Slide 5 of 22

Basic Network Components

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A network combines end devices, connecting devices, and links. These create or receive data, provide paths, and carry signals between devices.

- End devices create or receive data: computers, phones, servers, and printers.
- Connecting devices provide paths: a switch joins local wired devices, a wireless access point connects wireless devices, and a router joins networks.
- Links carry signals: Ethernet cables, optical fiber, and wireless radio connect compatible interfaces.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Discover everyday networking, classify network coverage and architectures, and follow a simple exchange between devices.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- End devices create or receive data: computers, phones, servers, and printers.
- Connecting devices provide paths: a switch joins local wired devices, a wireless access point connects wireless devices, and a router joins networks.
- Links carry signals: Ethernet cables, optical fiber, and wireless radio connect compatible interfaces.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of basic network components.

Key Takeaway: A network combines end devices, connecting devices, and links. These create or receive data, provide paths, and carry signals between devices.

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How Devices Connect

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Wired connections carry signals through a cable. Wireless connections carry signals over radio between compatible devices.

Interactive Demonstration: `introConnections` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Discover everyday networking, classify network coverage and architectures, and follow a simple exchange between devices.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Wired connections carry signals through a cable.
- Wireless connections carry signals over radio between compatible devices.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of how devices connect.

Key Takeaway: Wired connections carry signals through a cable. Wireless connections carry signals over radio between compatible devices.

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Local Network vs. Internet

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

A local network connects nearby devices so they can share data and resources. Internet access connects that network to services on other networks around the world.

Interactive Demonstration: `introConnectivity` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Discover everyday networking, classify network coverage and architectures, and follow a simple exchange between devices.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- A local network connects nearby devices so they can share data and resources.
- Internet access connects that network to services on other networks around the world.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of local network vs. internet.

Key Takeaway: A local network connects nearby devices so they can share data and resources. Internet access connects that network to services on other networks around the world.

Slide 8 of 22

Five Elements of Data Communication

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Data communication involves a sender, receiver, message, medium, and protocol. Together, these explain who exchanges data, what travels, how it travels, and which rules make it understandable.

- Sender: the device originating a message.
- Receiver: the intended destination of the message.
- Message: the data being exchanged.
- Medium: the physical or wireless path carrying the message.
- Protocol: rules for formatting, addressing, and exchanging the message. Both sides must support compatible rules.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Discover everyday networking, classify network coverage and architectures, and follow a simple exchange between devices.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Sender: the device originating a message.
- Receiver: the intended destination of the message.
- Message: the data being exchanged.
- Medium: the physical or wireless path carrying the message.
- Protocol: rules for formatting, addressing, and exchanging the message.
- Both sides must support compatible rules.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of five elements of data communication.

Key Takeaway: Data communication involves a sender, receiver, message, medium, and protocol. Together, these explain who exchanges data, what travels, how it travels, and which rules make it understandable.

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Example: one printer, five connected computers

EXAMPLE

📖 WHAT TO READ & DISPLAY ON SCREEN

Five computers can share one network printer through a switch. Each device has its own connection, allowing print requests and replies to travel through the local network.

- Computer 3 sends a document through the switch to the network printer.
- The computer is the sender, the printer is the receiver, and the print data is the message. Ethernet cables carry the exchange.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Discover everyday networking, classify network coverage and architectures, and follow a simple exchange between devices.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Computer 3 sends a document through the switch to the network printer.
- The computer is the sender, the printer is the receiver, and the print data is the message.
- Ethernet cables carry the exchange.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of example: one printer, five connected computers.

Key Takeaway: Five computers can share one network printer through a switch. Each device has its own connection, allowing print requests and replies to travel through the local network.

Slide 10 of 22

A Simple Data Exchange

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

A request asks another device to provide information or a service. When that device replies, it becomes the sender and the original requester becomes the receiver.

Interactive Demonstration: `introExchange` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Discover everyday networking, classify network coverage and architectures, and follow a simple exchange between devices.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- One computer requests a shared document and another responds.
- Sender and receiver roles change when the reply travels back.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of a simple data exchange.

Key Takeaway: A request asks another device to provide information or a service. When that device replies, it becomes the sender and the original requester becomes the receiver.

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Three Benefits of Networking

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Networking makes resources, information, and services available to multiple users. Sharing, collaboration, and managed backups help people work together more efficiently.

- Resource sharing: devices can use shared printers, storage, and an Internet connection.
- Collaboration: people exchange documents and work with shared information.
- Centralized backups: managed copies reduce scattered files and support recovery.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Discover everyday networking, classify network coverage and architectures, and follow a simple exchange between devices.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Resource sharing: devices can use shared printers, storage, and an Internet connection.
- Collaboration: people exchange documents and work with shared information.
- Centralized backups: managed copies reduce scattered files and support recovery.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of three benefits of networking.

Key Takeaway: Networking makes resources, information, and services available to multiple users. Sharing, collaboration, and managed backups help people work together more efficiently.

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Two Risks of Networking

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Shared access can expose information when permissions are poorly controlled. Shared equipment can also affect many users when it fails.

- Security exposure: an improperly exposed share can reveal data or permit unwanted changes.
- Shared failures: a failed central device or service can affect many connected users.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Discover everyday networking, classify network coverage and architectures, and follow a simple exchange between devices.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Security exposure: an improperly exposed share can reveal data or permit unwanted changes.
- Shared failures: a failed central device or service can affect many connected users.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of two risks of networking.

Key Takeaway: Shared access can expose information when permissions are poorly controlled. Shared equipment can also affect many users when it fails.

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Two Performance Measures: Bandwidth vs Latency

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Bandwidth describes how much data a link can carry per second. Latency describes delay, so a high-capacity connection can still feel slow when responses take time.

- Bandwidth: the capacity of a link, usually measured in bits per second.
- Latency: the time delay in communication. A high-capacity link may still have high delay.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Discover everyday networking, classify network coverage and architectures, and follow a simple exchange between devices.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Bandwidth: the capacity of a link, usually measured in bits per second.
- Latency: the time delay in communication.
- A high-capacity link may still have high delay.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of two performance measures: bandwidth vs latency.

Key Takeaway: Bandwidth describes how much data a link can carry per second. Latency describes delay, so a high-capacity connection can still feel slow when responses take time.

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Four network coverage types

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Network coverage describes the geographic scope of connected devices and sites. Personal, local, metropolitan, and wide area networks differ in scope rather than following universal distance cutoffs.

- Personal Area Network (PAN): devices around a person, such as a phone and Bluetooth headset.
- Local Area Network (LAN): a room, building, or local campus network, including a wireless Local Area Network (LAN).
- Metropolitan Area Network (MAN): interconnects sites across a metropolitan area.
- Wide Area Network (WAN): spans wider regions, often using carrier links. The Internet interconnects many networks.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Discover everyday networking, classify network coverage and architectures, and follow a simple exchange between devices.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Personal Area Network (PAN): devices around a person, such as a phone and Bluetooth headset.
- Local Area Network (LAN): a room, building, or local campus network, including a wireless Local Area Network (LAN).
- Metropolitan Area Network (MAN): interconnects sites across a metropolitan area.
- Wide Area Network (WAN): spans wider regions, often using carrier links.
- The Internet interconnects many networks.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of four network coverage types.

Key Takeaway: Network coverage describes the geographic scope of connected devices and sites. Personal, local, metropolitan, and wide area networks differ in scope rather than following universal distance cutoffs.

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Personal Area Network (PAN)

EXAMPLE

📖 WHAT TO READ & DISPLAY ON SCREEN

A Personal Area Network connects devices around one person. It usually covers a short range and can use Bluetooth or wired connections.

- The phone sends audio to the headset over Bluetooth.
- These devices form a network around one person; this does not require an Internet connection.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Discover everyday networking, classify network coverage and architectures, and follow a simple exchange between devices.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- The phone sends audio to the headset over Bluetooth.
- These devices form a network around one person; this does not require an Internet connection.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of personal area network (pan).

Key Takeaway: A Personal Area Network connects devices around one person. It usually covers a short range and can use Bluetooth or wired connections.

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Local Area Network (LAN)

EXAMPLE

📖 WHAT TO READ & DISPLAY ON SCREEN

A Local Area Network connects devices within a limited area, such as a room, building, or campus. It lets those devices exchange data and share resources through wired or wireless links.

- Each of the five computers and the printer has its own Ethernet cable to the switch: six occupied switch ports.
- The computers can send print jobs to the reachable printer. Internet access is unnecessary for this local exchange.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Discover everyday networking, classify network coverage and architectures, and follow a simple exchange between devices.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Each of the five computers and the printer has its own Ethernet cable to the switch: six occupied switch ports.
- The computers can send print jobs to the reachable printer.
- Internet access is unnecessary for this local exchange.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of local area network (lan).

Key Takeaway: A Local Area Network connects devices within a limited area, such as a room, building, or campus. It lets those devices exchange data and share resources through wired or wireless links.

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Metropolitan Area Network (MAN)

EXAMPLE

📖 WHAT TO READ & DISPLAY ON SCREEN

A Metropolitan Area Network connects networks across a city or metropolitan area. It can link sites such as library branches, municipal offices, or campuses within that area.

- A city-operated network interconnects library branches across a metropolitan area.
- Each branch also has its own local network; the city links connect those separate sites.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Discover everyday networking, classify network coverage and architectures, and follow a simple exchange between devices.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- A city-operated network interconnects library branches across a metropolitan area.
- Each branch also has its own local network; the city links connect those separate sites.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of metropolitan area network (man).

Key Takeaway: A Metropolitan Area Network connects networks across a city or metropolitan area. It can link sites such as library branches, municipal offices, or campuses within that area.

Slide 18 of 22

Wide Area Network (WAN)

EXAMPLE

📖 WHAT TO READ & DISPLAY ON SCREEN

A Wide Area Network connects networks across large geographic areas, such as regions or countries. Organizations commonly use provider services to link their distant sites.

- A provider service joins campuses in different regions. Routers connect the campus networks to that service.
- The wide-area link joins local networks; it does not replace their local switches or wireless access points.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Discover everyday networking, classify network coverage and architectures, and follow a simple exchange between devices.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- A provider service joins campuses in different regions.
- Routers connect the campus networks to that service.
- The wide-area link joins local networks; it does not replace their local switches or wireless access points.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of wide area network (wan).

Key Takeaway: A Wide Area Network connects networks across large geographic areas, such as regions or countries. Organizations commonly use provider services to link their distant sites.

Slide 19 of 22

Two network service architectures

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Network architecture describes which devices request and provide services. Peer-to-peer and client-server arrangements can both operate within the same geographic network type.

- Peer-to-peer: peers can both request and provide resources, such as one Personal Computer (PC) sharing a folder with another.
- Client-server: clients request services from dedicated service providers, such as a school file server.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Discover everyday networking, classify network coverage and architectures, and follow a simple exchange between devices.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Peer-to-peer: peers can both request and provide resources, such as one PC sharing a folder with another.
- Client-server: clients request services from dedicated service providers, such as a school file server.
- One Local Area Network (LAN) can use both architectures.
- A server is a role, not simply a large or expensive computer.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of two network service architectures.

Key Takeaway: Network architecture describes which devices request and provide services. Peer-to-peer and client-server arrangements can both operate within the same geographic network type.

Slide 20 of 22

P2P vs Client-Server Availability

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

A shared resource is available only while its host and required connection are working. Peer-hosted files depend on their individual owners, while centralized files depend on the server.

Interactive Demonstration: `availabilityDemo` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Discover everyday networking, classify network coverage and architectures, and follow a simple exchange between devices.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Compare the same shared files when a peer or the central server goes offline.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of p2p vs client-server availability.

Key Takeaway: A shared resource is available only while its host and required connection are working. Peer-hosted files depend on their individual owners, while centralized files depend on the server.

Slide 21 of 22

Two link choices: wired and wireless

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Wired links carry signals through a physical cable, while wireless links use radio. One network can combine both methods without changing its geographic classification.

- Ethernet provides a cabled link with predictable interference characteristics and a physical port requirement.
- Wi-Fi provides radio access and mobility; walls, interference, distance, and shared airtime affect performance.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Discover everyday networking, classify network coverage and architectures, and follow a simple exchange between devices.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Ethernet provides a cabled link with predictable interference characteristics and a physical port requirement.
- Wi-Fi provides radio access and mobility; walls, interference, distance, and shared airtime affect performance.
- Neither medium is automatically secure.
- Use authentication, encryption, and appropriate access controls.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of two link choices: wired and wireless.

Key Takeaway: Wired links carry signals through a physical cable, while wireless links use radio. One network can combine both methods without changing its geographic classification.

TOPIC 2 OF 13

Topic 2: Network Topologies and Basic Network Design

Build connections, compare layouts, and investigate what happens when a link fails.

Curriculum: Networking 1

Excluded: Title (Slide 1), Objectives (Slide 2), and Summary (Final Slide)

Discussion Units: 10 Slides

Slide 3 of 13

Two Views of Topology: Physical and Logical

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Network topology describes how devices and connections are arranged. Physical topology shows the actual layout, while logical topology describes how communication travels.

- Physical topology: which devices have direct connections and where the cables or wireless links are arranged.
- Logical topology: how communication uses those connections, including the path and access rules followed by the data.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Build connections, compare layouts, and investigate what happens when a link fails.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Physical topology: which devices have direct connections and where the cables or wireless links are arranged.
- Logical topology: how communication uses those connections, including the path and access rules followed by the data.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of two views of topology: physical and logical.

Key Takeaway: Network topology describes how devices and connections are arranged. Physical topology shows the actual layout, while logical topology describes how communication travels.

Slide 4 of 13

Six network topologies

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Star, bus, ring, mesh, tree, and hybrid networks use different connection patterns. Those patterns affect expansion, cost, available paths, and the impact of failures.

- Star: endpoints connect to a central switch. A single endpoint cable failure usually isolates that endpoint.
- Bus: a classic shared coaxial backbone requires correct termination. A backbone break disrupts the segment.
- Ring: a classic single ring passes traffic along a closed path. A break disrupts this simple model; protected rings may recover.
- Full mesh: every node connects directly to every other node; alternate paths can survive one link failure.
- Tree: branches connect hierarchically. An upstream link failure isolates the subtree below it.
- Hybrid: combines layouts, such as two stars joined by a backbone. The effect of a failure depends on where it occurs.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Build connections, compare layouts, and investigate what happens when a link fails.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Star: endpoints connect to a central switch.
- A single endpoint cable failure usually isolates that endpoint.
- Bus: a classic shared coaxial backbone requires correct termination.
- A backbone break disrupts the segment.
- Ring: a classic single ring passes traffic along a closed path.
- A break disrupts this simple model; protected rings may recover.
- Full mesh: every node connects directly to every other node; alternate paths can survive one link failure.
- Tree: branches connect hierarchically.
- An upstream link failure isolates the subtree below it.
- Hybrid: combines layouts, such as two stars joined by a backbone.
- The effect of a failure depends on where it occurs.
- Redundancy can improve availability but adds cost and control requirements.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of six network topologies.

Key Takeaway: Star, bus, ring, mesh, tree, and hybrid networks use different connection patterns. Those patterns affect expansion, cost, available paths, and the impact of failures.

Slide 5 of 13

Star Topology

EXAMPLE

📖 WHAT TO READ & DISPLAY ON SCREEN

A star connects each endpoint to a central device using its own link. This example has five computers and one printer, so the switch needs six endpoint connections.

- Five computers and one printer form six spokes around one switch.
- Unplugging Computer 2's cable isolates that computer. A failed central switch affects all six endpoints.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Build connections, compare layouts, and investigate what happens when a link fails.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Five computers and one printer form six spokes around one switch.
- Unplugging Computer 2's cable isolates that computer.
- A failed central switch affects all six endpoints.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of star topology.

Key Takeaway: A star connects each endpoint to a central device using its own link. This example has five computers and one printer, so the switch needs six endpoint connections.

Slide 6 of 13

Bus Topology

EXAMPLE

📖 WHAT TO READ & DISPLAY ON SCREEN

A bus connects devices to a shared backbone rather than to a central switch. Signals use the common cable, and correct termination is needed at its ends.

- Each computer attaches through a tap to the same backbone; computers do not forward traffic for their neighbors.
- This classic-bus model treats a broken backbone as disrupting the entire segment.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Build connections, compare layouts, and investigate what happens when a link fails.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Each computer attaches through a tap to the same backbone; computers do not forward traffic for their neighbors.
- This classic-bus model treats a broken backbone as disrupting the entire segment.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of bus topology.

Key Takeaway: A bus connects devices to a shared backbone rather than to a central switch. Signals use the common cable, and correct termination is needed at its ends.

Slide 7 of 13

Ring Topology

EXAMPLE

📖 WHAT TO READ & DISPLAY ON SCREEN

A ring connects each device to two neighbors to form a closed loop. This example uses an unprotected logical ring, so a break interrupts its circulation.

- The highlighted sequence follows neighboring connections around the loop.
- A protected ring can recover differently; this example deliberately has no protection path.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Build connections, compare layouts, and investigate what happens when a link fails.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- The highlighted sequence follows neighboring connections around the loop.
- A protected ring can recover differently; this example deliberately has no protection path.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of ring topology.

Key Takeaway: A ring connects each device to two neighbors to form a closed loop. This example uses an unprotected logical ring, so a break interrupts its circulation.

Slide 8 of 13

Mesh Topology

EXAMPLE

📖 WHAT TO READ & DISPLAY ON SCREEN

A full mesh connects every node directly to every other node. This provides alternate paths but requires more links as the network grows.

- Five fully interconnected nodes need ten distinct links.
- The path demonstration is an abstract routed graph; Ethernet loops require separate control mechanisms.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Build connections, compare layouts, and investigate what happens when a link fails.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Five fully interconnected nodes need ten distinct links.
- The path demonstration is an abstract routed graph; Ethernet loops require separate control mechanisms.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of mesh topology.

Key Takeaway: A full mesh connects every node directly to every other node. This provides alternate paths but requires more links as the network grows.

Slide 9 of 13

Tree Topology

EXAMPLE

📖 WHAT TO READ & DISPLAY ON SCREEN

A tree arranges connections in levels, with branches below a central root. An upstream link can affect every endpoint in the branch beneath it.

- Computer A reaches Computer B through Branch A, the core, and Branch B.
- Removing the core-to-Branch B link isolates Computer B while Branch A keeps its core connection.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Build connections, compare layouts, and investigate what happens when a link fails.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Computer A reaches Computer B through Branch A, the core, and Branch B.
- Removing the core-to-Branch B link isolates Computer B while Branch A keeps its core connection.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of tree topology.

Key Takeaway: A tree arranges connections in levels, with branches below a central root. An upstream link can affect every endpoint in the branch beneath it.

Slide 10 of 13 Hybrid Topology

EXAMPLE

📖 WHAT TO READ & DISPLAY ON SCREEN

A hybrid combines connection patterns within one network. This example joins two switched groups through one inter-switch link.

- Computers in different groups use both switches and the connecting backbone.
- If that backbone fails, the groups separate; devices on the same working switch can remain connected.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Build connections, compare layouts, and investigate what happens when a link fails.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Computers in different groups use both switches and the connecting backbone.
- If that backbone fails, the groups separate; devices on the same working switch can remain connected.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of hybrid topology.

Key Takeaway: A hybrid combines connection patterns within one network. This example joins two switched groups through one inter-switch link.

Slide 11 of 13 Handling Link and Device Failures

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

A failed cable or device can disconnect one endpoint, a branch, or the whole network segment. Test faults and trace available paths between devices across each topology layout.

Interactive Demonstration: `topology` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Build connections, compare layouts, and investigate what happens when a link fails.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Modeled connectivity • choose a layout, test cable/device failures, and trace communication paths.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of handling link and device failures.

Key Takeaway: A failed cable or device can disconnect one endpoint, a branch, or the whole network segment. Test faults and trace available paths between devices across each topology layout.

Slide 12 of 13

Choosing a Network Topology

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A useful topology meets the network's capacity, coverage, growth, and availability needs. A layout that is easy to expand may still depend on a central device or upstream connection.

- Count the computers, printers, servers, and uplinks that need connections; allow spare switch ports for growth.
- Choose suitable cable lengths and record which device uses each port. Clear labels make expansion and fault tracing easier.
- Consider the effect of one failed link or central device. Extra Ethernet links require appropriate loop prevention rather than simply connecting every available port.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Build connections, compare layouts, and investigate what happens when a link fails.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Count the computers, printers, servers, and uplinks that need connections; allow spare switch ports for growth.
- Choose suitable cable lengths and record which device uses each port.
- Clear labels make expansion and fault tracing easier.
- Consider the effect of one failed link or central device.
- Extra Ethernet links require appropriate loop prevention rather than simply connecting every available port.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of choosing a network topology.

Key Takeaway: A useful topology meets the network's capacity, coverage, growth, and availability needs. A layout that is easy to expand may still depend on a central device or upstream connection.

TOPIC 3 OF 13

Topic 3: Network Devices and Components

Identify the hardware and follow the different jobs of switching, routing, and wireless access.

Curriculum: Networking 1

Excluded: Title (Slide 1), Objectives (Slide 2), and Summary (Final Slide)

Discussion Units: 19 Slides

Slide 3 of 22

Nine common network devices

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Network devices connect hosts, carry signals, forward data, or enforce access rules. A single product can combine several functions, so its behavior matters more than its shape.

- Network Interface Card (NIC): provides a host network interface.
- Hub: repeats incoming bits to other ports without learning Media Access Control (MAC) destinations.
- Switch: learns Media Access Control (MAC) information and forwards local-link frames.
- Router: forwards Internet Protocol (IP) packets between networks using routes.
- Modem: adapts signals for a provider access technology.
- Wireless access point: bridges Wi-Fi stations onto a Local Area Network (LAN).
- Repeater: regenerates or retransmits signals.
- Bridge: joins link-layer segments; a switch is a multiport bridge.
- Firewall: permits or blocks traffic according to policy.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *The office has too few wired ports and weak Wi-Fi in a meeting room.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- NIC: provides a host network interface.
- Hub: repeats incoming bits to other ports without learning MAC destinations.
- Switch: learns MAC information and forwards local-link frames.
- Router: forwards IP packets between networks using routes.
- Modem: adapts signals for a provider access technology.
- Wireless access point: bridges Wi-Fi stations onto a LAN.
- Repeater: regenerates or retransmits signals.
- Bridge: joins link-layer segments; a switch is a multiport bridge.
- Firewall: permits or blocks traffic according to policy.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of nine common network devices.

Key Takeaway: Network devices connect hosts, carry signals, forward data, or enforce access rules. A single product can combine several functions, so its behavior matters more than its shape.

Slide 4 of 22

Network Interface Card (NIC)

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A Network Interface Card provides the physical connection and hardware Media Access Control (MAC) address that allows a computer to send and receive frames on a local link.

- Physical connection: Houses the physical port (such as Registered Jack 45 (RJ45) or Wi-Fi radio) and controller chip for host communication.
- Unique hardware identity: Permanently assigned a factory-burned 48-bit Media Access Control (MAC) address for local link delivery.
- Data transmission: Converts host memory data into electrical or radio bitstreams, framing data with Media Access Control (MAC) addresses and CRC checksums.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *The office has too few wired ports and weak Wi-Fi in a meeting room.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Physical connection: Houses the physical port (such as RJ45 or Wi-Fi radio) and controller chip for host communication.
- Unique hardware identity: Permanently assigned a factory-burned 48-bit Media Access Control (MAC) address for local link delivery.
- Data transmission: Converts host memory data into electrical or radio bitstreams, framing data with MAC addresses and CRC checksums.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of network interface card (nic).

Key Takeaway: A Network Interface Card provides the physical connection and hardware Media Access Control (MAC) address that allows a computer to send and receive frames on a local link.

Slide 5 of 22

Network Hub

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A network hub is a legacy Layer 1 multiport repeater that blindly broadcasts incoming electrical signals out all other ports, creating a shared collision domain.

- Multiport repeater: Connects multiple twisted-pair Ethernet cables in a single shared electrical bus.
- No address awareness: Operates purely at Layer 1; it cannot read Media Access Control (MAC) addresses, Internet Protocol (IP) packets, or inspect frames.
- Broadcast behavior: Any electrical bit received on one port is broadcast out to every other port, creating a single shared collision domain.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *The office has too few wired ports and weak Wi-Fi in a meeting room.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Multiport repeater: Connects multiple twisted-pair Ethernet cables in a single shared electrical bus.
- No address awareness: Operates purely at Layer 1; it cannot read MAC addresses, IP packets, or inspect frames.
- Broadcast behavior: Any electrical bit received on one port is broadcast out to every other port, creating a single shared collision domain.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of network hub.

Key Takeaway: A network hub is a legacy Layer 1 multiport repeater that blindly broadcasts incoming electrical signals out all other ports, creating a shared collision domain.

Slide 6 of 22 **Network Switch** CONCEPT

📖 **WHAT TO READ & DISPLAY ON SCREEN**

A network switch is an intelligent Layer 2 device that learns Media Access Control (MAC) addresses on each port to forward frames directly to their intended destination.

- Dedicated bandwidth: Connects local hosts with dedicated point-to-point links, eliminating collisions between ports.
- Media Access Control (MAC) address table: Dynamically inspects the source MAC of incoming frames and maps each address to its incoming physical port.
- Forwarding decision: When a frame arrives, the switch forwards known unicast directly to the target port; only unknown destinations are flooded.

💡 **BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)**

🌟 **PRACTICAL CONTEXT / INTUITION**

> *The office has too few wired ports and weak Wi-Fi in a meeting room.*

🗣️ **HOW TO EXPLAIN IT (TALKING POINTS)**

- Dedicated bandwidth: Connects local hosts with dedicated point-to-point links, eliminating collisions between ports.
- MAC address table: Dynamically inspects the source MAC of incoming frames and maps each address to its incoming physical port.
- Forwarding decision: When a frame arrives, the switch forwards known unicast directly to the target port; only unknown destinations are flooded.

❓ **QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)**

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of network switch.

Key Takeaway: A network switch is an intelligent Layer 2 device that learns Media Access Control (MAC) addresses on each port to forward frames directly to their intended destination.

Slide 7 of 22

Network Router

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A router is a Layer 3 internetworking device that uses routing tables to forward Internet Protocol (IP) packets across different networks and broadcast domains.

- Network boundary: Separates distinct broadcast domains and connects local subnets to WANs or the global Internet.
- Routing table: Maintains a table of network destination prefixes, next-hop Internet Protocol (IP) addresses, and outgoing interfaces.
- Packet forwarding: Reads destination Internet Protocol (IP) headers, decrements the Time to Live (TTL), determines the best path, rewrites the Layer 2 frame, and forwards.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *The office has too few wired ports and weak Wi-Fi in a meeting room.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Network boundary: Separates distinct broadcast domains and connects local subnets to WANs or the global Internet.
- Routing table: Maintains a table of network destination prefixes, next-hop IP addresses, and outgoing interfaces.
- Packet forwarding: Reads destination IP headers, decrements the TTL, determines the best path, rewrites the Layer 2 frame, and forwards.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of network router.

Key Takeaway: A router is a Layer 3 internetworking device that uses routing tables to forward Internet Protocol (IP) packets across different networks and broadcast domains.

Slide 8 of 22

Modem (Modulator-Demodulator)

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A modem converts digital computer pulses into analog signals for provider transmission lines, adapting local Ethernet to external provider media.

- Media adaptation: Bridges home or office Ethernet to provider infrastructures like telephone lines (Digital Subscriber Line (DSL)), cable (DOCSIS), or fiber (ONT).
- Modulation & demodulation: Modulates outgoing digital square waves into analog carrier frequencies and demodulates incoming analog waves back to bits.
- Signal conversion: Local Area Network (LAN) Ethernet packets enter the modem, are modulated onto high-frequency RF carriers over coaxial or copper cables, and reach the ISP headend.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *The office has too few wired ports and weak Wi-Fi in a meeting room.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Media adaptation: Bridges home or office Ethernet to provider infrastructures like telephone lines (DSL), cable (DOCSIS), or fiber (ONT).
- Modulation & demodulation: Modulates outgoing digital square waves into analog carrier frequencies and demodulates incoming analog waves back to bits.
- Signal conversion: LAN Ethernet packets enter the modem, are modulated onto high-frequency RF carriers over coaxial or copper cables, and reach the ISP headend.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of modem (modulator-demodulator).

Key Takeaway: A modem converts digital computer pulses into analog signals for provider transmission lines, adapting local Ethernet to external provider media.

Slide 9 of 22

Wireless Access Point (AP)

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A wireless access point bridges radio frequency signals from Wi-Fi client stations onto a wired Ethernet local area network.

- Radio transceiver: Transmits and receives radio waves across 2.4 GHz, 5 GHz, or 6 GHz frequency bands.
- Layer 2 bridge: Bridges IEEE 802.11 wireless frames into IEEE 802.3 wired Ethernet frames without changing Internet Protocol (IP) addresses.
- Client bridging: Broadcasts an Service Set Identifier (SSID) beacon, authenticates wireless client devices, and transparently forwards their traffic to the wired switch via PoE uplink.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *The office has too few wired ports and weak Wi-Fi in a meeting room.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Radio transceiver: Transmits and receives radio waves across 2.4 GHz, 5 GHz, or 6 GHz frequency bands.
- Layer 2 bridge: Bridges IEEE 802.11 wireless frames into IEEE 802.3 wired Ethernet frames without changing IP addresses.
- Client bridging: Broadcasts an SSID beacon, authenticates wireless client devices, and transparently forwards their traffic to the wired switch via PoE uplink.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of wireless access point (ap).

Key Takeaway: A wireless access point bridges radio frequency signals from Wi-Fi client stations onto a wired Ethernet local area network.

Slide 10 of 22 **Network Repeater** CONCEPT

📖 **WHAT TO READ & DISPLAY ON SCREEN**

A repeater regenerates and retimes attenuated signals at the physical layer, extending the effective reach of a network cable or wireless link.

- Overcoming attenuation: Counteracts signal loss and cable resistance that degrade data beyond standard maximum run lengths.
- Signal regeneration: Rather than merely amplifying noise, a digital repeater reconstructs clean, full-voltage digital pulses.
- Signal restoration: Listens on an input interface, cleans and retimes the incoming waveform, and retransmits it at full strength across the extended medium segment.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *The office has too few wired ports and weak Wi-Fi in a meeting room.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Overcoming attenuation: Counteracts signal loss and cable resistance that degrade data beyond standard maximum run lengths.
- Signal regeneration: Rather than merely amplifying noise, a digital repeater reconstructs clean, full-voltage digital pulses.
- Signal restoration: Listens on an input interface, cleans and retimes the incoming waveform, and retransmits it at full strength across the extended medium segment.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of network repeater.

Key Takeaway: A repeater regenerates and retimes attenuated signals at the physical layer, extending the effective reach of a network cable or wireless link.

Slide 11 of 22

Network Bridge

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A network bridge joins two physical Local Area Network (LAN) segments at Layer 2, filtering local traffic to isolate collisions while forwarding necessary inter-segment frames.

- Traffic segmentation: Divides a busy network into two smaller collision domains while keeping them in the same subnet.
- Filtering table: Maintains a list of Media Access Control (MAC) addresses on Segment A and Segment B to decide whether traffic needs to cross.
- Frame filtering: If sender and receiver are on Segment A, the bridge filters (drops) the frame from crossing; if the target is on Segment B, it forwards it across.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *The office has too few wired ports and weak Wi-Fi in a meeting room.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Traffic segmentation: Divides a busy network into two smaller collision domains while keeping them in the same subnet.
- Filtering table: Maintains a list of MAC addresses on Segment A and Segment B to decide whether traffic needs to cross.
- Frame filtering: If sender and receiver are on Segment A, the bridge filters (drops) the frame from crossing; if the target is on Segment B, it forwards it across.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of network bridge.

Key Takeaway: A network bridge joins two physical Local Area Network (LAN) segments at Layer 2, filtering local traffic to isolate collisions while forwarding necessary inter-segment frames.

Slide 12 of 22 **Network Firewall** CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A firewall enforces security policies at network boundaries, inspecting packet headers and connection states to permit authorized traffic and block threats.

- Security perimeter: Sits at network boundaries (e.g. between Local Area Network (LAN) and Internet) or as host software to defend against unauthorized access.
- Inspection methods: Uses packet filtering (Internet Protocol (IP)/ports), stateful inspection (tracking active connections), or deep packet inspection (DPI).
- Policy enforcement: Inspects packet headers against security rules; permitted traffic (e.g. outbound web) is passed, while unauthorized packets are dropped or rejected.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *The office has too few wired ports and weak Wi-Fi in a meeting room.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Security perimeter: Sits at network boundaries (e.g.
- between LAN and Internet) or as host software to defend against unauthorized access.
- Inspection methods: Uses packet filtering (IP/ports), stateful inspection (tracking active connections), or deep packet inspection (DPI).
- Policy enforcement: Inspects packet headers against security rules; permitted traffic (e.g.
- outbound web) is passed, while unauthorized packets are dropped or rejected.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of network firewall.

Key Takeaway: A firewall enforces security policies at network boundaries, inspecting packet headers and connection states to permit authorized traffic and block threats.

Slide 13 of 22

How Interfaces and Switches Connect Local Devices

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A network interface lets a host send and receive data on a link. A switch connects local links and uses learned hardware addresses to forward Ethernet frames.

- A hub repeats incoming bits out its other ports; it does not learn destination Media Access Control (MAC) addresses.
- A switch learns source Media Access Control (MAC) addresses and forwards known unicast frames toward the learned destination port. Unknown destinations are flooded within the Virtual Local Area Network (VLAN).
- A bridge joins Local Area Network (LAN) segments at layer 2. A switch is effectively a multiport bridge.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *The office has too few wired ports and weak Wi-Fi in a meeting room.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- A hub repeats incoming bits out its other ports; it does not learn destination MAC addresses.
- A switch learns source MAC addresses and forwards known unicast frames toward the learned destination port.
- Unknown destinations are flooded within the VLAN.
- A bridge joins LAN segments at layer 2.
- A switch is effectively a multiport bridge.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of how interfaces and switches connect local devices.

Key Takeaway: A network interface lets a host send and receive data on a link. A switch connects local links and uses learned hardware addresses to forward Ethernet frames.

Slide 14 of 22

How Switches Learn MAC Addresses

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

A switch learns a frame's source Media Access Control address on the port where that frame arrives. Known destinations use learned ports, while unknown destinations require flooding to eligible ports.

Interactive Demonstration: `macLearningDemo` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Identify the hardware and follow the different jobs of switching, routing, and wireless access.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Inject frames between three modeled ports and observe flooding, learning, and directed forwarding.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of how switches learn mac addresses.

Key Takeaway: A switch learns a frame's source Media Access Control address on the port where that frame arrives. Known destinations use learned ports, while unknown destinations require flooding to eligible ports.

Slide 15 of 22

Routers, Access Points, and Modems

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A router forwards packets between networks, while an access point connects wireless devices to a local network. A modem or optical terminal supports the provider's access technology.

- A modem adapts signals for a provider access technology, such as cable or Digital Subscriber Line (DSL). An optical network terminal serves a fiber access role.
- A wireless access point bridges Wi-Fi clients onto a Local Area Network (LAN). An AP alone is not necessarily a router.
- A repeater regenerates or retransmits signals. It does not choose Internet Protocol (IP) routes. Home gateways often combine router, switch, Access Point (AP), Dynamic Host Configuration Protocol (DHCP), and firewall functions.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *The office has too few wired ports and weak Wi-Fi in a meeting room.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- A modem adapts signals for a provider access technology, such as cable or DSL.
- An optical network terminal serves a fiber access role.
- A wireless access point bridges Wi-Fi clients onto a LAN.
- An AP alone is not necessarily a router.
- A repeater regenerates or retransmits signals.
- It does not choose IP routes.
- Home gateways often combine router, switch, AP, DHCP, and firewall functions.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of routers, access points, and modems.

Key Takeaway: A router forwards packets between networks, while an access point connects wireless devices to a local network. A modem or optical terminal supports the provider's access technology.

Slide 16 of 22

Example: Switch vs Router Roles

EXAMPLE

📖 WHAT TO READ & DISPLAY ON SCREEN

A local printer and a remote website require different delivery paths. The switch connects local devices, while the router handles forwarding toward another network.

- Local print traffic travels from the computer through the switch to the printer.
- Traffic to a remote network goes through the switch to the router, which chooses an appropriate next hop.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Identify the hardware and follow the different jobs of switching, routing, and wireless access.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Local print traffic travels from the computer through the switch to the printer.
- Traffic to a remote network goes through the switch to the router, which chooses an appropriate next hop.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of example: switch vs router roles.

Key Takeaway: A local printer and a remote website require different delivery paths. The switch connects local devices, while the router handles forwarding toward another network.

Slide 17 of 22

Interfaces, Link Lights, and Firewalls

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

An interface connects a device to a network, while a firewall controls permitted traffic. A working link indicator confirms only part of the connection and does not prove Internet or application access.

- Local Area Network (LAN) ports usually connect local clients; a Wide Area Network (WAN) interface connects toward another network/provider. Check the actual model's labels.
- A firewall permits or blocks traffic according to policy. It is not a guarantee that all allowed traffic is safe.
- Link lights show physical/link status, not proof of correct Internet Protocol (IP) settings, Domain Name System (DNS), or Internet service.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *The office has too few wired ports and weak Wi-Fi in a meeting room.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- LAN ports usually connect local clients; a WAN interface connects toward another network/provider.
- Check the actual model's labels.
- A firewall permits or blocks traffic according to policy.
- It is not a guarantee that all allowed traffic is safe.
- Link lights show physical/link status, not proof of correct IP settings, DNS, or Internet service.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of interfaces, link lights, and firewalls.

Key Takeaway: An interface connects a device to a network, while a firewall controls permitted traffic. A working link indicator confirms only part of the connection and does not prove Internet or application access.

Slide 18 of 22

VLAN Concepts and Routing Principles

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Virtual LANs divide a single physical switch into isolated logical broadcast domains, while routers connect them using static or dynamic routing rules.

- Virtual Local Area Network (LAN) (Virtual Local Area Network (VLAN)): Divides one physical switch into multiple separate logical broadcast domains (e.g., VLAN 10 for Students, VLAN 20 for Faculty).
- Broadcast Isolation: Broadcast traffic (such as Address Resolution Protocol (ARP) and Dynamic Host Configuration Protocol (DHCP)) stays confined inside its assigned Virtual Local Area Network (VLAN), boosting performance and security.
- Inter-VLAN Routing: Because devices in different VLANs cannot communicate directly, all inter-VLAN traffic must pass through a router or Layer 3 switch.
- Static Routing: A network administrator manually enters fixed destination routes into the router table (simple, predictable, low overhead for small networks).
- Dynamic Routing: Routers automatically discover paths and exchange network routes using protocols like OSPF and RIP (automatically updates if a link fails).

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

☀️ PRACTICAL CONTEXT / INTUITION

> *The office has too few wired ports and weak Wi-Fi in a meeting room.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Virtual LAN (VLAN): Divides one physical switch into multiple separate logical broadcast domains (e.g., VLAN 10 for Students, VLAN 20 for Faculty).
- Broadcast Isolation: Broadcast traffic (such as ARP and DHCP) stays confined inside its assigned VLAN, boosting performance and security.
- Inter-VLAN Routing: Because devices in different VLANs cannot communicate directly, all inter-VLAN traffic must pass through a router or Layer 3 switch.
- Static Routing: A network administrator manually enters fixed destination routes into the router table (simple, predictable, low overhead for small networks).
- Dynamic Routing: Routers automatically discover paths and exchange network routes using protocols like OSPF and RIP (automatically updates if a link fails).
- VLANs replace the need to buy separate physical switches for each school department, saving hardware costs while enforcing security.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of vlan concepts and routing principles.

Key Takeaway: Virtual LANs divide a single physical switch into isolated logical broadcast domains, while routers connect them using static or dynamic routing rules.

Slide 19 of 22

Network Device Functions Explorer

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Device functions explain how traffic moves through a network. A hub repeats incoming bits, a switch forwards local frames, and a router forwards packets between networks.

Interactive Demonstration: `devices` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *The office has too few wired ports and weak Wi-Fi in a meeting room.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Conceptual device gallery • inspect labeled interfaces and compare a hub, switch, and router.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of network device functions explorer.

Key Takeaway: Device functions explain how traffic moves through a network. A hub repeats incoming bits, a switch forwards local frames, and a router forwards packets between networks.

Slide 20 of 22

Expand a small office

APPLICATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Additional wired endpoints need enough switch ports, while weak wireless coverage may need a better-placed access point. Adding the right function is more useful than adding another box without checking the existing network.

- Add a suitable switch for Ethernet ports and an Access Point (AP) with a wired uplink for wireless coverage.
- Use a router for traffic between networks, rather than buying another hub.
- Check existing gateway features before adding duplicate Dynamic Host Configuration Protocol (DHCP) servers.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *The office has too few wired ports and weak Wi-Fi in a meeting room.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Add a suitable switch for Ethernet ports and an AP with a wired uplink for wireless coverage.
- Use a router for traffic between networks, rather than buying another hub.
- Check existing gateway features before adding duplicate DHCP servers.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of expand a small office.

Key Takeaway: Additional wired endpoints need enough switch ports, while weak wireless coverage may need a better-placed access point. Adding the right function is more useful than adding another box without checking the existing network.

Slide 21 of 22

Device Roles Along a Network Path

WALKTHROUGH

📖 WHAT TO READ & DISPLAY ON SCREEN

A request can pass through several devices, each performing a different job. Interfaces, switching, routing, and provider access work together without being interchangeable functions.

- The Personal Computer (PC) Network Interface Card (NIC) sends a frame through an Ethernet port to a switch. Link indicators show link detection, not proof of Internet Protocol (IP) or Internet access.
- The switch forwards a known local unicast toward its learned destination port. A hub would repeat incoming bits to other ports.
- For a remote Internet Protocol (IP) network, a router selects the next hop. Its Local Area Network (LAN) and Wide Area Network (WAN) interfaces face different network roles.
- A home gateway may also contain an Access Point (AP), switch, Dynamic Host Configuration Protocol (DHCP) service, Network Address Translation (NAT), and firewall. A modem or optical terminal serves the provider access technology.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *The office has too few wired ports and weak Wi-Fi in a meeting room.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- The PC NIC sends a frame through an Ethernet port to a switch.
- Link indicators show link detection, not proof of IP or Internet access.
- The switch forwards a known local unicast toward its learned destination port.
- A hub would repeat incoming bits to other ports.
- For a remote IP network, a router selects the next hop.
- Its LAN and WAN interfaces face different network roles.
- A home gateway may also contain an AP, switch, DHCP service, NAT, and firewall.
- A modem or optical terminal serves the provider access technology.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of device roles along a network path.

Key Takeaway: A request can pass through several devices, each performing a different job. Interfaces, switching, routing, and provider access work together without being interchangeable functions.

TOPIC 4 OF 13

Topic 4: Open Systems Interconnection (OSI) Reference Model and Network Communication

Explore the seven communication layers, follow data encapsulation, and compare OSI with the TCP/IP model.

Curriculum: Networking 1

Excluded: Title (Slide 1), Objectives (Slide 2), and Summary (Final Slide)

Discussion Units: 14 Slides

Slide 3 of 17

What Is the OSI Reference Model?

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Open Systems Interconnection is a seven-layer reference model for describing communication responsibilities. It helps explain how network functions cooperate without being a protocol that sends data itself.

Interactive Demonstration: `osiIntroduction` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

A user sends a message; the exchange depends on signals, framing, routing, transport, and an available application service.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- It helps learning, design discussions, and troubleshooting.
- It is not itself a network protocol or a mandatory seven-step program.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of what is the osi reference model?.

Key Takeaway: Open Systems Interconnection is a seven-layer reference model for describing communication responsibilities. It helps explain how network functions cooperate without being a protocol that sends data itself.

Slide 4 of 17

Seven OSI layer responsibilities

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

The Open Systems Interconnection model names seven responsibilities involved in communication. A classic memory aid from Layer 1 to 7 is "Please Do Not Throw Sausage Pizza Away".

- Application: Network services and protocols used directly by end-user applications.
- Presentation: Data formatting, character encoding, translation, and encryption.
- Session: Dialog management, establishing, maintaining, and synchronizing communication sessions.
- Transport: End-to-end communication, segmentation, flow control, and error recovery.
- Network: Logical addressing and path determination (routing) across interconnected networks.
- Data Link: Physical addressing, framing, media access, and local hop delivery.
- Physical: Transmission and reception of raw binary bitstreams over a physical medium.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A user sends a message; the exchange depends on signals, framing, routing, transport, and an available application service.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Application: Network services and protocols used directly by end-user applications.
- Presentation: Data formatting, character encoding, translation, and encryption.
- Session: Dialog management, establishing, maintaining, and synchronizing communication sessions.
- Transport: End-to-end communication, segmentation, flow control, and error recovery.
- Network: Logical addressing and path determination (routing) across interconnected networks.
- Data Link: Physical addressing, framing, media access, and local hop delivery.
- Physical: Transmission and reception of raw binary bitstreams over a physical medium.
- Each layer provides services to the layer above it and relies on the layer beneath it.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of seven osi layer responsibilities.

Key Takeaway: The Open Systems Interconnection model names seven responsibilities involved in communication. A classic memory aid from Layer 1 to 7 is "Please Do Not Throw Sausage Pizza Away".

Slide 5 of 17

What Is Encapsulation?

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Encapsulation adds control information around application data for transport, network, and link delivery. The receiver interprets the corresponding headers and framing to recover the data.

- Application data is carried in a Transmission Control Protocol (TCP) segment or User Datagram Protocol (UDP) datagram, then an Internet Protocol (IP) packet, then a link-layer frame. Physical signals carry the frame bits.
- The receiver interprets and removes the relevant headers. Ethernet also uses a trailer for error detection.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A user sends a message; the exchange depends on signals, framing, routing, transport, and an available application service.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Application data is carried in a TCP segment or UDP datagram, then an IP packet, then a link-layer frame.
- Physical signals carry the frame bits.
- The receiver interprets and removes the relevant headers.
- Ethernet also uses a trailer for error detection.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of what is encapsulation?.

Key Takeaway: Encapsulation adds control information around application data for transport, network, and link delivery. The receiver interprets the corresponding headers and framing to recover the data.

Slide 6 of 17 Data Encapsulation and Delivery

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Messages use different delivery information at different communication layers. Transport behavior and the local-link path explain how the same application data reaches its receiver.

Interactive Demonstration: `osi` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

A user sends a message; the exchange depends on signals, framing, routing, transport, and an available application service.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Conceptual walkthrough • select any OSI layer, step through encapsulation, and compare TCP/UDP loss handling.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of data encapsulation and delivery.

Key Takeaway: Messages use different delivery information at different communication layers. Transport behavior and the local-link path explain how the same application data reaches its receiver.

Slide 7 of 17 Layer 7: Application

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

The Application layer describes the network services used by applications. Its protocols define the requests and responses exchanged between them.

Interactive Demonstration: `osiLayer` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

A user sends a message; the exchange depends on signals, framing, routing, transport, and an available application service.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- A browser uses HTTP to request a resource.
- The layer describes network services, not the browser's graphical interface.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of layer 7: application.

Key Takeaway: The Application layer describes the network services used by applications. Its protocols define the requests and responses exchanged between them.

Slide 8 of 17

Layer 6: Presentation

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

The Presentation layer formats, encrypts, and compresses data so communicating devices understand each other. It handles character sets (like UTF-8) and secure SSL/Transport Layer Security (TLS) encryption.

Interactive Demonstration: `osiLayer` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

A user sends a message; the exchange depends on signals, framing, routing, transport, and an available application service.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Both peers need compatible representations.
- Encryption and compression illustrate these responsibilities, but actual TCP/IP software need not expose a separate Presentation layer.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of layer 6: presentation.

Key Takeaway: The Presentation layer formats, encrypts, and compresses data so communicating devices understand each other. It handles character sets (like UTF-8) and secure SSL/Transport Layer Security (TLS) encryption.

Slide 9 of 17

Layer 5: Session

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

The Session layer establishes, manages, and terminates ongoing conversations between applications. It keeps dialogue organized and ensures open communication channels do not overlap.

Interactive Demonstration: `osiLayer` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

A user sends a message; the exchange depends on signals, framing, routing, transport, and an available application service.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Think of a coordinated conversation.
- A web login session and a TCP connection are different concepts; TCP connection setup is a Transport function, not exclusively Session.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of layer 5: session.

Key Takeaway: The Session layer establishes, manages, and terminates ongoing conversations between applications. It keeps dialogue organized and ensures open communication channels do not overlap.

Slide 10 of 17

Layer 4: Transport

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

The Transport layer delivers data between applications on communicating devices. The transport protocol determines whether missing data is recovered and delivered in order.

Interactive Demonstration: `osiLayer` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

A user sends a message; the exchange depends on signals, framing, routing, transport, and an available application service.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- TCP supplies an ordered reliable byte stream using sequence numbers, acknowledgments, and retransmission.
- UDP itself does not guarantee delivery or ordering.
- Applications can build reliability on UDP.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of layer 4: transport.

Key Takeaway: The Transport layer delivers data between applications on communicating devices. The transport protocol determines whether missing data is recovered and delivered in order.

Slide 11 of 17

Two transport services: TCP and UDP

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Transmission Control Protocol supplies an ordered reliable byte stream, while User Datagram Protocol supplies independent datagrams. Applications choose the behavior their communication needs rather than assuming one transport is always better.

- Transmission Control Protocol (TCP): an ordered reliable byte stream with sequence numbers, acknowledgments, and retransmission.
- User Datagram Protocol (UDP): datagrams without built-in delivery or ordering guarantees; applications may add reliability.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A user sends a message; the exchange depends on signals, framing, routing, transport, and an available application service.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- TCP: an ordered reliable byte stream with sequence numbers, acknowledgments, and retransmission.
- UDP: datagrams without built-in delivery or ordering guarantees; applications may add reliability.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of two transport services: tcp and udp.

Key Takeaway: Transmission Control Protocol supplies an ordered reliable byte stream, while User Datagram Protocol supplies independent datagrams. Applications choose the behavior their communication needs rather than assuming one transport is always better.

Slide 12 of 17 **Layer 3: Network** SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

The Network layer handles addressing and delivery between different networks. Routers use destination addresses to choose where packets go next.

Interactive Demonstration: `osiLayer` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A user sends a message; the exchange depends on signals, framing, routing, transport, and an available application service.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Routers choose a next hop using a routing table.
- A packet keeps its remote destination IP across this no-NAT example; each link uses its own framing.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of layer 3: network.

Key Takeaway: The Network layer handles addressing and delivery between different networks. Routers use destination addresses to choose where packets go next.

Slide 13 of 17

Layer 2: Data Link

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

The Data Link layer carries frames across one local link at a time. Link addresses identify where a frame should go on that link.

Interactive Demonstration: `osiLayer` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

A user sends a message; the exchange depends on signals, framing, routing, transport, and an available application service.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- An Ethernet switch learns source MAC addresses.
- A known destination is forwarded to its learned port; an unknown unicast is flooded to other eligible ports in the VLAN.
- Error detection is not automatic end-to-end correction.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of layer 2: data link.

Key Takeaway: The Data Link layer carries frames across one local link at a time. Link addresses identify where a frame should go on that link.

Slide 14 of 17

Layer 1: Physical

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

The Physical layer carries bits as signals through a medium. Those signals can travel through copper, fiber, or radio.

Interactive Demonstration: `osiLayer` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

A user sends a message; the exchange depends on signals, framing, routing, transport, and an available application service.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Cables, connectors, and transceivers carry signals.
- Wi-Fi involves both Physical radio transmission and Data Link framing/media access.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of layer 1: physical.

Key Takeaway: The Physical layer carries bits as signals through a medium. Those signals can travel through copper, fiber, or radio.

Slide 15 of 17

Four layers in the TCP/IP model

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

The common Transmission Control Protocol (TCP)/Internet Protocol (IP) model groups communication into Application, Transport, Internet, and Link. These groups connect application services to the delivery and transmission functions underneath them.

- Application: application protocols plus representation and dialog functions.
- Transport: process-to-process services, including Transmission Control Protocol (TCP) and User Datagram Protocol (UDP).
- Internet: Internet Protocol (IP) logical addressing and forwarding across networks.
- Link: local framing, media access, and physical transmission.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A user sends a message; the exchange depends on signals, framing, routing, transport, and an available application service.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Application: application protocols plus representation and dialog functions.
- Transport: process-to-process services, including TCP and UDP.
- Internet: IP logical addressing and forwarding across networks.
- Link: local framing, media access, and physical transmission.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of four layers in the tcp/ip model.

Key Takeaway: The common Transmission Control Protocol (TCP)/Internet Protocol (IP) model groups communication into Application, Transport, Internet, and Link. These groups connect application services to the delivery and transmission functions underneath them.

Slide 16 of 17

Comparing the OSI and TCP/IP Models

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

The seven-layer Open Systems Interconnection model and the common four-layer Transmission Control Protocol (TCP)/Internet Protocol (IP) model group communication responsibilities differently. Their mapping is useful for explanation but does not impose identical software boundaries.

Interactive Demonstration: `osiComparison` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A user sends a message; the exchange depends on signals, framing, routing, transport, and an available application service.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- OSI 7–5 map to Application, 4 to Transport, 3 to Internet, and 2–1 to Network Access / Link.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of comparing the osi and tcp/ip models.

Key Takeaway: The seven-layer Open Systems Interconnection model and the common four-layer Transmission Control Protocol (TCP)/Internet Protocol (IP) model group communication responsibilities differently. Their mapping is useful for explanation but does not impose identical software boundaries.

TOPIC 5 OF 13

Topic 5: Network Cable Types and Media

Compare network cable types, categories, fiber optics, coaxial, and wireless media.

Curriculum: Networking 1 **Excluded:** Title (Slide 1), Objectives (Slide 2), and Summary (Final Slide) **Discussion Units:** 8 Slides

Slide 3 of 11

Five network media choices

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A transmission medium carries signals between network interfaces. Twisted-pair copper, fiber, coaxial cable, and radio differ in construction, reach, interference behavior, and installation needs.

- Unshielded Twisted Pair (UTP): Four unshielded twisted copper pairs, no overall shield. Twisting reduces electromagnetic interference (EMI).
- Shielded Twisted Pair (STP): Shielded twisted-pair cable with foil or braid shielding; requires proper grounding and bonding.
- Fiber: Carries light signals through glass or plastic cores; immune to EMI and spans long distances.
- Coaxial: Copper core with dielectric insulator and braided metallic shield; used for broadband and legacy bus.
- Wireless: Radio frequency transmission providing mobility across shared, unguided electromagnetic airspace.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A 180 m outdoor path joins two buildings with electrical equipment nearby.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- UTP: Four unshielded twisted copper pairs, no overall shield.
- Twisting reduces electromagnetic interference (EMI).
- STP: Shielded twisted-pair cable with foil or braid shielding; requires proper grounding and bonding.
- Fiber: Carries light signals through glass or plastic cores; immune to EMI and spans long distances.
- Coaxial: Copper core with dielectric insulator and braided metallic shield; used for broadband and legacy bus.
- Wireless: Radio frequency transmission providing mobility across shared, unguided electromagnetic airspace.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of five network media choices.

Key Takeaway: A transmission medium carries signals between network interfaces. Twisted-pair copper, fiber, coaxial cable, and radio differ in construction, reach, interference behavior, and installation needs.

Slide 4 of 11

Three Ethernet cable categories

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A cable category describes specified performance characteristics for compatible Ethernet installations. Supported rate and reach also depend on the standard, connectors, channel length, and installation quality.

- Cat 5e: Supports 1 Gb/s (1000BASE-T) over a standard 100 m channel.
- Cat 6: Supports 1 Gb/s at 100 m; supports 10 Gb/s (10GBASE-T) up to 55 m.
- Cat 6A: Supports 10 Gb/s (10GBASE-T) over a full 100 m channel.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A 180 m outdoor path joins two buildings with electrical equipment nearby.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Cat 5e: Supports 1 Gb/s (1000BASE-T) over a standard 100 m channel.
- Cat 6: Supports 1 Gb/s at 100 m; supports 10 Gb/s (10GBASE-T) up to 55 m.
- Cat 6A: Supports 10 Gb/s (10GBASE-T) over a full 100 m channel.
- A standard channel comprises up to 90 m of permanent horizontal link plus 10 m of patch cords.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of three ethernet cable categories.

Key Takeaway: A cable category describes specified performance characteristics for compatible Ethernet installations. Supported rate and reach also depend on the standard, connectors, channel length, and installation quality.

Slide 5 of 11

Fiber-Optic Cable Construction and Types

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Optical fiber carries light pulses through glass or plastic cores instead of electrical signals. Single-mode and multi-mode fiber provide immune, high-speed, long-distance communication.

- Core and Cladding: Ultra-thin glass core surrounded by reflective cladding that bounces light along the fiber using total internal reflection.
- Single-Mode Fiber (SMF): Extremely thin 9-micron core using laser light; spans dozens of kilometers for campus backbones and telco WANs.
- Multi-Mode Fiber (MMF): Wider 50 or 62.5-micron core using LED or VCSEL light; spans up to 550 m for high-speed data center links.
- Immunity to Interference: Because light does not conduct electricity, fiber is immune to electrical noise, motors, and lightning surges.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌞 PRACTICAL CONTEXT / INTUITION

> *A 180 m outdoor path joins two buildings with electrical equipment nearby.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Core and Cladding: Ultra-thin glass core surrounded by reflective cladding that bounces light along the fiber using total internal reflection.
- Single-Mode Fiber (SMF): Extremely thin 9-micron core using laser light; spans dozens of kilometers for campus backbones and telco WANs.
- Multi-Mode Fiber (MMF): Wider 50 or 62.5-micron core using LED or VCSEL light; spans up to 550 m for high-speed data center links.
- Immunity to Interference: Because light does not conduct electricity, fiber is immune to electrical noise, motors, and lightning surges.
- Never look directly into an active fiber optic connector or transceiver; invisible laser light can cause permanent retinal damage.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of fiber-optic cable construction and types.

Key Takeaway: Optical fiber carries light pulses through glass or plastic cores instead of electrical signals. Single-mode and multi-mode fiber provide immune, high-speed, long-distance communication.

Slide 6 of 11

Coaxial Cable Construction and Shielding

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Coaxial cable uses a concentric four-layer design that provides physical durability and electrical shielding. It is widely used in cable broadband modems and legacy bus architectures.

- Center Conductor: Solid copper or copper-clad steel wire that carries the high-frequency radio frequency signal.
- Dielectric Insulator: Non-conductive plastic foam layer that maintains precise physical spacing between center conductor and shield.
- Braided Metallic Shield: Woven copper or aluminum foil that blocks external electrical interference and completes the circuit ground.
- Common Use Cases: Modern cable broadband modems (DOCSIS), closed-circuit TV (CCTV), and legacy 10BASE2/10BASE5 bus networks.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A 180 m outdoor path joins two buildings with electrical equipment nearby.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Center Conductor: Solid copper or copper-clad steel wire that carries the high-frequency radio frequency signal.
- Dielectric Insulator: Non-conductive plastic foam layer that maintains precise physical spacing between center conductor and shield.
- Braided Metallic Shield: Woven copper or aluminum foil that blocks external electrical interference and completes the circuit ground.
- Common Use Cases: Modern cable broadband modems (DOCSIS), closed-circuit TV (CCTV), and legacy 10BASE2/10BASE5 bus networks.
- Coaxial cables terminate with threaded F-type connectors (for cable modems) or twist-lock BNC connectors (for legacy gear).

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of coaxial cable construction and shielding.

Key Takeaway: Coaxial cable uses a concentric four-layer design that provides physical durability and electrical shielding. It is widely used in cable broadband modems and legacy bus architectures.

Slide 7 of 11

Unguided Media: Wireless Radio Frequencies

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Unguided media transmit electromagnetic radio waves through the open air without physical cables. Wireless provides endpoint mobility across shared radio spectrum.

- Radio Frequencies (RF): Transmits digital data modulated onto microwave and radio carrier frequencies between 2.4 GHz and 6 GHz.
- Shared Transmission Airspace: Unlike dedicated copper links, all wireless stations in an area share the same radio frequency spectrum.
- Environmental Attenuation: Radio waves are absorbed, reflected, and scattered by concrete walls, metal filing cabinets, and water.
- Convenience vs. Stability: Wireless enables laptop and smartphone mobility, while wired Ethernet delivers guaranteed speed and zero interference.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A 180 m outdoor path joins two buildings with electrical equipment nearby.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Radio Frequencies (RF): Transmits digital data modulated onto microwave and radio carrier frequencies between 2.4 GHz and 6 GHz.
- Shared Transmission Airspace: Unlike dedicated copper links, all wireless stations in an area share the same radio frequency spectrum.
- Environmental Attenuation: Radio waves are absorbed, reflected, and scattered by concrete walls, metal filing cabinets, and water.
- Convenience vs.
- Stability: Wireless enables laptop and smartphone mobility, while wired Ethernet delivers guaranteed speed and zero interference.
- In enterprise networks, wireless access points are always connected back to the core network using wired Category 6 Ethernet cables.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of unguided media: wireless radio frequencies.

Key Takeaway: Unguided media transmit electromagnetic radio waves through the open air without physical cables. Wireless provides endpoint mobility across shared radio spectrum.

Slide 8 of 11

Transmission Media Comparison Workbench

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Different physical media solve different speed, distance, and interference challenges. Select media types to inspect cross-sections and identify optimal deployment scenarios.

Interactive Demonstration: `media` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

A 180 m outdoor path joins two buildings with electrical equipment nearby.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Illustrated comparison • inspect cable cross-sections and choose media for a task.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of transmission media comparison workbench.

Key Takeaway: Different physical media solve different speed, distance, and interference challenges. Select media types to inspect cross-sections and identify optimal deployment scenarios.

Slide 9 of 11

Campus Inter-Building Media Selection

APPLICATION

📖 WHAT TO READ & DISPLAY ON SCREEN

A standard 100 m copper Ethernet channel cannot span outdoor campus runs or withstand ground potential differences. Optical fiber provides electrical isolation and multi-kilometer reach.

- A standard 100 m copper Ethernet channel is insufficient for this direct path.
- A suitable fiber link with appropriate transceivers can meet reach and isolation requirements.
- Installation needs rated materials and professional planning; fiber is not simply a longer Registered Jack 45 (RJ45) cable.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A 180 m outdoor path joins two buildings with electrical equipment nearby.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- A standard 100 m copper Ethernet channel is insufficient for this direct path.
- A suitable fiber link with appropriate transceivers can meet reach and isolation requirements.
- Installation needs rated materials and professional planning; fiber is not simply a longer RJ45 cable.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of campus inter-building media selection.

Key Takeaway: A standard 100 m copper Ethernet channel cannot span outdoor campus runs or withstand ground potential differences. Optical fiber provides electrical isolation and multi-kilometer reach.

Slide 10 of 11

Campus Media Selection Walkthrough

WALKTHROUGH

📖 WHAT TO READ & DISPLAY ON SCREEN

Plan campus transmission links by evaluating channel distance, interference conditions, and transceiver requirements. Always verify physical cable ratings before deployment.

- Identify the required channel distance: horizontal floor runs under 90 m can use Category 6 twisted pair, while outdoor building runs over 100 m require optical fiber.
- Compare interference conditions: factory floors or conduits near high-voltage lines require fiber or properly bonded Shielded Twisted Pair (STP) cable.
- Select optical fiber profiles: single-mode fiber for long-distance inter-building backbones, and multi-mode fiber for high-speed switch-to-switch links within a data center.
- Verify physical termination and transceiver compatibility before applying power or configuring network interfaces.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A 180 m outdoor path joins two buildings with electrical equipment nearby.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Identify the required channel distance: horizontal floor runs under 90 m can use Category 6 twisted pair, while outdoor building runs over 100 m require optical fiber.
- Compare interference conditions: factory floors or conduits near high-voltage lines require fiber or properly bonded STP cable.
- Select optical fiber profiles: single-mode fiber for long-distance inter-building backbones, and multi-mode fiber for high-speed switch-to-switch links within a data center.
- Verify physical termination and transceiver compatibility before applying power or configuring network interfaces.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of campus media selection walkthrough.

Key Takeaway: Plan campus transmission links by evaluating channel distance, interference conditions, and transceiver requirements. Always verify physical cable ratings before deployment.

TOPIC 6 OF 13

Topic 6: Registered Jack 45 (RJ45) Crimping, Color Coding, and Cable Testing

Learn cabling tools, standardize pin color orders, terminate RJ45 plugs and keystone jacks, and test wire continuity.

Curriculum: Networking 1

Excluded: Title (Slide 1), Objectives (Slide 2), and Summary (Final Slide)

Discussion Units: 12 Slides

Slide 3 of 15

RJ45 Plugs and Keystone Jacks

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Twisted-pair Ethernet uses Eight Positions, Eight Contacts (8P8C) modular plugs for patch cords and 110 Insulation-Displacement Contact (IDC) keystone jacks for permanent wall runs. Both components must match the cable category and wiring standard.

- Registered Jack 45 (RJ45) (Eight Positions, Eight Contacts (8P8C)) plug: Modular male plug with eight gold-plated contacts, used to terminate flexible stranded patch cables.
- Keystone Jack: Modular female jack with 110 Insulation-Displacement Contact (IDC) contacts, mounted in wall faceplates and patch panels for permanent solid-core horizontal cabling.
- Category matching: Plugs and jacks must match the cable category and conductor gauge to prevent signal degradation.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌞 PRACTICAL CONTEXT / INTUITION

> *One PC has an intermittent link after a cable was sharply pulled from its desk.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- RJ45 (8P8C) plug: Modular male plug with eight gold-plated contacts, used to terminate flexible stranded patch cables.
- Keystone Jack: Modular female jack with 110 IDC contacts, mounted in wall faceplates and patch panels for permanent solid-core horizontal cabling.
- Category matching: Plugs and jacks must match the cable category and conductor gauge to prevent signal degradation.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of rj45 plugs and keystone jacks.

Key Takeaway: Twisted-pair Ethernet uses Eight Positions, Eight Contacts (8P8C) modular plugs for patch cords and 110 Insulation-Displacement Contact (IDC) keystone jacks for permanent wall runs. Both components must match the cable category and wiring standard.

Slide 4 of 15

Four cabling tool and tester functions

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Cabling tools prepare, terminate, or test different parts of an installation. Stripping, crimping, punch-down termination, and continuity testing each require the appropriate tool.

- Cable stripper: removes jacket without nicking conductor insulation.
- Crimping tool: terminates compatible modular plugs.
- Punch-down tool: seats conductors in compatible Insulation-Displacement Contact (IDC) jacks or patch panels.
- Basic Local Area Network (LAN) tester: checks conductor continuity and wire mapping.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *One PC has an intermittent link after a cable was sharply pulled from its desk.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Cable stripper: removes jacket without nicking conductor insulation.
- Crimping tool: terminates compatible modular plugs.
- Punch-down tool: seats conductors in compatible IDC jacks or patch panels.
- Basic LAN tester: checks conductor continuity and wire mapping.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of four cabling tool and tester functions.

Key Takeaway: Cabling tools prepare, terminate, or test different parts of an installation. Stripping, crimping, punch-down termination, and continuity testing each require the appropriate tool.

Slide 5 of 15

RJ45 Cable Crimping Tools

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Terminating an Eight Positions, Eight Contacts (8P8C) modular plug requires preparing dedicated cutting, stripping, crimping, and testing tools alongside quality Cat 5e/Cat 6 media.

- Registered Jack 45 (RJ45) Modular Crimper: Heavy-duty ratcheting tool with precision Eight Positions, Eight Contacts (8P8C) dies to compress contacts into copper conductors and engage the jacket strain relief.
- Unshielded Twisted Pair (UTP) Cable Stripper: Compact rotary wire stripper calibrated to cleanly score and remove the outer PVC jacket without nicking conductor insulation.
- Modular Registered Jack 45 (RJ45) Plugs: Cat 5e or Cat 6 clear polycarbonate male Eight Positions, Eight Contacts (8P8C) plugs featuring eight gold-plated contact pins and a flexible locking tab.
- Unshielded Twisted Pair (UTP) Cable: Four-pair 24 AWG stranded copper Ethernet cable designed for flexible patch cords between workstations and wall jacks.
- Flush Cutters / Snips: Precision diagonal cutting pliers used to trim untwisted, aligned wires straight and square across before insertion into the plug.
- Local Area Network (LAN) Cable Tester: Dual-unit continuity wire-map tester to confirm all eight pin paths and detect open circuits, short circuits, or crossed pairs.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *One PC has an intermittent link after a cable was sharply pulled from its desk.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- RJ45 Modular Crimper: Heavy-duty ratcheting tool with precision 8P8C dies to compress contacts into copper conductors and engage the jacket strain relief.
- UTP Cable Stripper: Compact rotary wire stripper calibrated to cleanly score and remove the outer PVC jacket without nicking conductor insulation.
- Modular RJ45 Plugs: Cat 5e or Cat 6 clear polycarbonate male 8P8C plugs featuring eight gold-plated contact pins and a flexible locking tab.
- Unshielded Twisted Pair (UTP) Cable: Four-pair 24 AWG stranded copper Ethernet cable designed for flexible patch cords between workstations and wall jacks.
- Flush Cutters / Snips: Precision diagonal cutting pliers used to trim untwisted, aligned wires straight and square across before insertion into the plug.
- LAN Cable Tester: Dual-unit continuity wire-map tester to confirm all eight pin paths and detect open circuits, short circuits, or crossed pairs.
- Always prepare spare 8P8C plugs; modular plugs cannot be reused once crimped.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of rj45 cable crimping tools.

Key Takeaway: Terminating an Eight Positions, Eight Contacts (8P8C) modular plug requires preparing dedicated cutting, stripping, crimping, and testing tools alongside quality Cat 5e/Cat 6 media.

Slide 6 of 15

Keystone Jack Punch-Down Tools

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Terminating a permanent wall jack requires specialized insulation displacement tools, punch-down supports, and rated solid-core cabling.

- 110 Impact Punch-Down Tool: Spring-loaded impact driver with a reversible 110/66 blade that seats conductors into Insulation-Displacement Contact (IDC) V-notches and cuts excess wire flush.
- Keystone Jack (110 Insulation-Displacement Contact (IDC)): Registered Jack 45 (RJ45) female modular jack with 110 insulation displacement contacts, universal T568A/T568B color coding, and snap-in faceplate clips.
- Unshielded Twisted Pair (UTP) Cable Stripper: Round jacket stripper used to remove 1 to 1.5 inches of outer jacket while preserving pair twists up to the Insulation-Displacement Contact (IDC) blocks.
- Solid-Core Unshielded Twisted Pair (UTP) Cable: Four-pair 23/24 AWG solid copper horizontal cable engineered for structured in-wall and ceiling cable runs.
- Punch-Down Palm Tool / Stand: Contoured palm puck or base holder that cradles the jack securely to prevent hand injury and ensure even downward punching pressure.
- Insulation-Displacement Contact (IDC) Dust Caps & Cable Tester: Snap-on protective dust covers to lock conductors firmly in place, plus a Local Area Network (LAN) tester with patch cord to verify end-to-end pin mapping.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *One PC has an intermittent link after a cable was sharply pulled from its desk.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- 110 Impact Punch-Down Tool: Spring-loaded impact driver with a reversible 110/66 blade that seats conductors into IDC V-notches and cuts excess wire flush.
- Keystone Jack (110 IDC): RJ45 female modular jack with 110 insulation displacement contacts, universal T568A/T568B color coding, and snap-in faceplate clips.
- UTP Cable Stripper: Round jacket stripper used to remove 1 to 1.5 inches of outer jacket while preserving pair twists up to the IDC blocks.
- Solid-Core UTP Cable: Four-pair 23/24 AWG solid copper horizontal cable engineered for structured in-wall and ceiling cable runs.
- Punch-Down Palm Tool / Stand: Contoured palm puck or base holder that cradles the jack securely to prevent hand injury and ensure even downward punching pressure.
- IDC Dust Caps & Cable Tester: Snap-on protective dust covers to lock conductors firmly in place, plus a LAN tester with patch cord to verify end-to-end pin mapping.
- Always orient the punch-down tool with the CUT blade facing outward toward the excess wire tail.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of keystone jack punch-down tools.

Key Takeaway: Terminating a permanent wall jack requires specialized insulation displacement tools, punch-down supports, and rated solid-core cabling.

Slide 7 of 15

T568A: eight conductor positions

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

T568A specifies which conductor occupies each of the eight contact positions. Read the color order from pin 1 to pin 8 using the stated connector orientation.

- Pin 1: White/Green.
- Pin 2: Green.
- Pin 3: White/Orange.
- Pin 4: Blue.
- Pin 5: White/Blue.
- Pin 6: Orange.
- Pin 7: White/Brown.
- Pin 8: Brown.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *One PC has an intermittent link after a cable was sharply pulled from its desk.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Pin 1: White/Green.
- Pin 2: Green.
- Pin 3: White/Orange.
- Pin 4: Blue.
- Pin 5: White/Blue.
- Pin 6: Orange.
- Pin 7: White/Brown.
- Pin 8: Brown.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of t568a: eight conductor positions.

Key Takeaway: T568A specifies which conductor occupies each of the eight contact positions. Read the color order from pin 1 to pin 8 using the stated connector orientation.

Slide 8 of 15

T568B: eight conductor positions

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

T568B is another standardized eight-contact conductor arrangement. It exchanges the orange and green pair positions relative to T568A while preserving the blue and brown positions.

- Pin 1: White/Orange.
- Pin 2: Orange.
- Pin 3: White/Green.
- Pin 4: Blue.
- Pin 5: White/Blue.
- Pin 6: Green.
- Pin 7: White/Brown.
- Pin 8: Brown.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *One PC has an intermittent link after a cable was sharply pulled from its desk.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Pin 1: White/Orange.
- Pin 2: Orange.
- Pin 3: White/Green.
- Pin 4: Blue.
- Pin 5: White/Blue.
- Pin 6: Green.
- Pin 7: White/Brown.
- Pin 8: Brown.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of t568b: eight conductor positions.

Key Takeaway: T568B is another standardized eight-contact conductor arrangement. It exchanges the orange and green pair positions relative to T568A while preserving the blue and brown positions.

Slide 9 of 15

Straight-Through vs Crossover Cables

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A straight-through cable uses the same termination standard at both ends. A traditional crossover changes selected pair positions, although many modern interfaces can adjust automatically.

- A conventional A–B crossover swaps pins 1↔3 and 2↔6 for traditional 10/100BASE-T transmit/receive pairs.
- Many modern interfaces support auto-MDI/Medium Dependent Interface Crossover (MDIX), so a crossover cable is often unnecessary. Gigabit Ethernet uses all four pairs.
- An equal but nonstandard color order may pass a simple continuity check while creating split pairs and poor performance.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *One PC has an intermittent link after a cable was sharply pulled from its desk.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- A conventional A–B crossover swaps pins 1↔3 and 2↔6 for traditional 10/100BASE-T transmit/receive pairs.
- Many modern interfaces support auto-MDI/MDIX, so a crossover cable is often unnecessary.
- Gigabit Ethernet uses all four pairs.
- An equal but nonstandard color order may pass a simple continuity check while creating split pairs and poor performance.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of straight-through vs crossover cables.

Key Takeaway: A straight-through cable uses the same termination standard at both ends. A traditional crossover changes selected pair positions, although many modern interfaces can adjust automatically.

Slide 10 of 15

Example: a straight-through patch cable

EXAMPLE

📖 WHAT TO READ & DISPLAY ON SCREEN

A straight-through patch cord connects unlike devices, such as a computer to an Ethernet switch. Both cable ends share identical pin assignments verified on a wire-map tester.

- A straight-through patch cable connects unlike devices, such as a computer to a switch.
- Both ends share identical pin assignments. A continuity tester verifies that each conductor reaches the same pin number on both ends.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Learn cabling tools, standardize pin color orders, terminate RJ45 plugs and keystone jacks, and test wire continuity.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- A straight-through patch cable connects unlike devices, such as a computer to a switch.
- Both ends share identical pin assignments.
- A continuity tester verifies that each conductor reaches the same pin number on both ends.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of example: a straight-through patch cable.

Key Takeaway: A straight-through patch cord connects unlike devices, such as a computer to an Ethernet switch. Both cable ends share identical pin assignments verified on a wire-map tester.

Slide 11 of 15

Three common wire-map faults

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A wire map identifies which contact paths connect between cable ends. Opens, shorts, and miswires describe different continuity problems and require different corrections.

- Open: a conductor has no continuous end-to-end path.
- Short: conductors are accidentally joined.
- Miswire: a conductor arrives at an unexpected pin.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *One PC has an intermittent link after a cable was sharply pulled from its desk.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Open: a conductor has no continuous end-to-end path.
- Short: conductors are accidentally joined.
- Miswire: a conductor arrives at an unexpected pin.
- A wire-map pass does not certify pair integrity, signal quality, or cable bandwidth.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of three common wire-map faults.

Key Takeaway: A wire map identifies which contact paths connect between cable ends. Opens, shorts, and miswires describe different continuity problems and require different corrections.

Slide 12 of 15

8P8C Modular Cable Termination Workbench

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Terminating twisted-pair cable requires aligning conductors to standard pin positions. Modeled testing calculates wire-map continuity and flags common termination faults.

Interactive Demonstration: `cable` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *One PC has an intermittent link after a cable was sharply pulled from its desk.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Modeled wire-map tester • select all eight conductors at both ends.
- Tester results are calculated from your mappings.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of 8p8c modular cable termination workbench.

Key Takeaway: Terminating twisted-pair cable requires aligning conductors to standard pin positions. Modeled testing calculates wire-map continuity and flags common termination faults.

Slide 13 of 15

Patch Cable Troubleshooting and Repair

APPLICATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Physical damage to patch cables commonly causes intermittent link drops or reduced link speeds. Inspect strain relief and terminate a new plug rather than altering Internet Protocol (IP) settings.

- Check connector strain relief, bent contacts, and damaged cable before changing Internet Protocol (IP) settings.
- Replace a damaged plug with a compatible one, preserving the chosen standard.
- Record wire-map results and test the actual link; continuity alone cannot prove rate or reliability.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *One PC has an intermittent link after a cable was sharply pulled from its desk.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Check connector strain relief, bent contacts, and damaged cable before changing IP settings.
- Replace a damaged plug with a compatible one, preserving the chosen standard.
- Record wire-map results and test the actual link; continuity alone cannot prove rate or reliability.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of patch cable troubleshooting and repair.

Key Takeaway: Physical damage to patch cables commonly causes intermittent link drops or reduced link speeds. Inspect strain relief and terminate a new plug rather than altering Internet Protocol (IP) settings.

Slide 14 of 15

Cable Termination and Testing Walkthrough

WALKTHROUGH

📖 WHAT TO READ & DISPLAY ON SCREEN

Follow a structured termination workflow from jacket stripping and conductor alignment to full insertion and crimping. Always verify continuity with a Local Area Network (LAN) cable tester before deployment.

- Use the contact-side view with the latch behind and cable entering downward. Pin 1 is left; confirm this orientation at both ends.
- Cut cleanly and strip only the required jacket without nicking insulation. Arrange the selected T568 standard and preserve twists close to termination.
- Trim conductors evenly, insert them fully into a compatible plug, and confirm the jacket is under strain relief.
- Inspect all eight contact positions before crimping. Keep fingers away from blades and jaws; a bad crimp usually requires a new plug.
- Connect main and remote tester units. Trace all eight paths, repair opens/shorts/miswires, and check the actual authorized link separately.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌞 PRACTICAL CONTEXT / INTUITION

> *One PC has an intermittent link after a cable was sharply pulled from its desk.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Use the contact-side view with the latch behind and cable entering downward.
- Pin 1 is left; confirm this orientation at both ends.
- Cut cleanly and strip only the required jacket without nicking insulation.
- Arrange the selected T568 standard and preserve twists close to termination.
- Trim conductors evenly, insert them fully into a compatible plug, and confirm the jacket is under strain relief.
- Inspect all eight contact positions before crimping.
- Keep fingers away from blades and jaws; a bad crimp usually requires a new plug.
- Connect main and remote tester units.
- Trace all eight paths, repair opens/shorts/miswires, and check the actual authorized link separately.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of cable termination and testing walkthrough.

Key Takeaway: Follow a structured termination workflow from jacket stripping and conductor alignment to full insertion and crimping. Always verify continuity with a Local Area Network (LAN) cable tester before deployment.

TOPIC 7 OF 13

Topic 7: Internet Protocol (IP) Addressing and Subnet Basics

Understand IPv4 address structure, decimal octets, classes, private ranges, subnet boundaries, and IPv6 fundamentals.

Curriculum: Networking 1

Excluded: Title (Slide 1), Objectives (Slide 2), and Summary (Final Slide)

Discussion Units: 15 Slides

Slide 3 of 18

What Is an Internet Protocol Address?

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

An Internet Protocol (IP) address is a unique logical Layer 3 numerical identifier assigned to every device on a network. While a physical Media Access Control (MAC) address identifies hardware on a single local cable, an IP address is hierarchical and location-based, enabling routers to direct data packets across interconnected local, enterprise, and global networks.

- Logical identity: Assigned by network configuration, unlike a permanent physical Media Access Control (MAC) address.
- Hierarchical routing: Routers use the network portion to direct traffic across interconnected networks.
- Octet ranges: Each of the four decimal octets ranges from 0 to 255 (representing 8 binary bits each).

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom needs addresses for 28 PCs plus a router interface.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Logical identity: Assigned by network configuration, unlike a permanent physical MAC address.
- Hierarchical routing: Routers use the network portion to direct traffic across interconnected networks.
- Octet ranges: Each of the four decimal octets ranges from 0 to 255 (representing 8 binary bits each).

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of what is an internet protocol address?.

Key Takeaway: An Internet Protocol (IP) address is a unique logical Layer 3 numerical identifier assigned to every device on a network. While a physical Media Access Control (MAC) address identifies hardware on a single local cable, an IP address is hierarchical and location-based, enabling routers to direct data packets across interconnected local, enterprise, and global networks.

Slide 4 of 18

Two Parts of an IP Address: Network and Host

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Every Internet Protocol (IP) address is divided into two parts: the network portion identifies the network, and the host portion identifies the specific device.

- Network portion: Identifies the specific network or subnet.
- Host portion: Identifies the individual device on that network.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom needs addresses for 28 PCs plus a router interface.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Network portion: Identifies the specific network or subnet.
- Host portion: Identifies the individual device on that network.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of two parts of an ip address: network and host.

Key Takeaway: Every Internet Protocol (IP) address is divided into two parts: the network portion identifies the network, and the host portion identifies the specific device.

Slide 5 of 18

What Is a Subnet and Why Do We Subnet?

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A subnet is a smaller, organized group of computers within a network, such as separating Students, Staff, and Guest Wi-Fi.

- Organization: Divides a large network into manageable, logical departments.
- Traffic control: Keeps broadcast messages inside one group so they do not slow down other computers.
- Security boundaries: Devices in different subnets cannot talk directly; they must pass through a router or firewall where security rules apply.
- Address efficiency: Allows network administrators to assign addresses without wasting large blocks.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom needs addresses for 28 PCs plus a router interface.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Organization: Divides a large network into manageable, logical departments.
- Traffic control: Keeps broadcast messages inside one group so they do not slow down other computers.
- Security boundaries: Devices in different subnets cannot talk directly; they must pass through a router or firewall where security rules apply.
- Address efficiency: Allows network administrators to assign addresses without wasting large blocks.
- Subnetting is a logical division created by the Subnet Mask; it does not require buying new physical switches.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of what is a subnet and why do we subnet?.

Key Takeaway: A subnet is a smaller, organized group of computers within a network, such as separating Students, Staff, and Guest Wi-Fi.

Slide 6 of 18

The Subnet Mask: The Dividing Boundary

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A subnet mask is a 32-bit filter that marks the boundary between network bits and host bits using consecutive ones and zeros.

- Network boundary: Consecutive binary 1s identify the network portion; consecutive binary 0s identify the host portion.
- Dotted-decimal format: A mask with 24 ones is written as 255.255.255.0.
- Classless Inter-Domain Routing (CIDR) prefix notation: Counting the 24 network ones is written shorthand as /24 (pronounced 'slash twenty-four').
- Universal /24 standard: The most common mask in small offices, schools, and homes, providing up to 254 usable computers.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom needs addresses for 28 PCs plus a router interface.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Network boundary: Consecutive binary 1s identify the network portion; consecutive binary 0s identify the host portion.
- Dotted-decimal format: A mask with 24 ones is written as 255.255.255.0.
- CIDR prefix notation: Counting the 24 network ones is written shorthand as /24 (pronounced 'slash twenty-four').
- Universal /24 standard: The most common mask in small offices, schools, and homes, providing up to 254 usable computers.
- A subnet mask must always have contiguous 1s followed by contiguous 0s.
- A mask like 255.255.0.255 is invalid.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of the subnet mask: the dividing boundary.

Key Takeaway: A subnet mask is a 32-bit filter that marks the boundary between network bits and host bits using consecutive ones and zeros.

Slide 7 of 18

How an Address and Mask Identify the Network

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

A bitwise AND combines address bits with subnet-mask bits to obtain the network address. Mask ones preserve network bits, while mask zeros clear the host portion.

Interactive Demonstration: `bitwiseDemo` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Understand IPv4 address structure, decimal octets, classes, private ranges, subnet boundaries, and IPv6 fundamentals.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Display address bits, mask bits, and the actual bitwise network result.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of how an address and mask identify the network.

Key Takeaway: A bitwise AND combines address bits with subnet-mask bits to obtain the network address. Mask ones preserve network bits, while mask zeros clear the host portion.

Slide 8 of 18

Interpreting an IPv4 Host Configuration

WALKTHROUGH

📖 WHAT TO READ & DISPLAY ON SCREEN

A host configuration combines an address, network boundary, and optional service or routing settings. Understanding each field prevents confusing name resolution with local or remote delivery.

- In 192.168.10.25/24, the prefix marks 24 network bits and eight host bits. The mask is 255.255.255.0.
- The gateway 192.168.10.1 is another usable address on that local subnet and handles destinations without a more specific route.
- A Domain Name System (DNS) resolver maps names to records. It may equal the gateway when that gateway forwards DNS, but this is not required.
- An octet uses weights 128, 64, 32, 16, 8, 4, 2, 1; adding enabled weights converts eight bits into a decimal value.
- Private addresses are not globally routed; loopback, link-local, shared carrier, documentation, multicast, and reserved ranges also need separate treatment.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom needs addresses for 28 PCs plus a router interface.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- In 192.168.10.25/24, the prefix marks 24 network bits and eight host bits.
- The mask is 255.255.255.0.
- The gateway 192.168.10.1 is another usable address on that local subnet and handles destinations without a more specific route.
- A DNS resolver maps names to records.
- It may equal the gateway when that gateway forwards DNS, but this is not required.
- An octet uses weights 128, 64, 32, 16, 8, 4, 2, 1; adding enabled weights converts eight bits into a decimal value.
- Private addresses are not globally routed; loopback, link-local, shared carrier, documentation, multicast, and reserved ranges also need separate treatment.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of interpreting an ipv4 host configuration.

Key Takeaway: A host configuration combines an address, network boundary, and optional service or routing settings. Understanding each field prevents confusing name resolution with local or remote delivery.

Slide 9 of 18

Inside Every Subnet: Network, Broadcast, and Host IPs

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Every subnet is simply an Internet Protocol (IP) address group with a starting address, an ending address, and usable host addresses in between.

- Network ID (The First Internet Protocol (IP)): The very first address (e.g., 192.168.1.0). It names the entire subnet and cannot be assigned to a computer.
- Broadcast Internet Protocol (IP) (The Last IP): The very last address (e.g., 192.168.1.255). It sends messages to every device in the subnet at once.
- Usable Host IPs (The Middle Numbers): All addresses between the first and last (e.g., 192.168.1.1 through 192.168.1.254). Assigned to PCs, phones, and printers.
- Total Minus 2 Rule: Because the first (Network ID) and last (Broadcast Internet Protocol (IP)) are reserved, usable host count is always Total IPs − 2.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom needs addresses for 28 PCs plus a router interface.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Network ID (The First IP): The very first address (e.g., 192.168.1.0).
- It names the entire subnet and cannot be assigned to a computer.
- Broadcast IP (The Last IP): The very last address (e.g., 192.168.1.255).
- It sends messages to every device in the subnet at once.
- Usable Host IPs (The Middle Numbers): All addresses between the first and last (e.g., 192.168.1.1 through 192.168.1.254).
- Assigned to PCs, phones, and printers.
- Total Minus 2 Rule: Because the first (Network ID) and last (Broadcast IP) are reserved, usable host count is always Total IPs − 2.
- In a /24 subnet with 256 total addresses, subtracting 2 gives exactly 254 usable host addresses.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of inside every subnet: network, broadcast, and host ips.

Key Takeaway: Every subnet is simply an Internet Protocol (IP) address group with a starting address, an ending address, and usable host addresses in between.

Slide 10 of 18

Subnet Calculator Demo

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

The address and prefix determine a subnet's mask, boundary, and address count. Network, broadcast, and usable-host conventions must then be applied appropriately.

Interactive Demonstration: subnet — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

A classroom needs addresses for 28 PCs plus a router interface.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Real calculation • any valid IPv4 address and prefix, binary boundary, guided steps, and explained calculations.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of subnet calculator demo.

Key Takeaway: The address and prefix determine a subnet's mask, boundary, and address count. Network, broadcast, and usable-host conventions must then be applied appropriately.

Slide 11 of 18

CIDR Prefix Notation (/24 Shortcut)

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Classless Inter-Domain Routing (CIDR) prefix notation states the number of consecutive network ones in the mask using a slash (such as /24), replacing cumbersome dotted-decimal masks.

- /24 means 24 ones in the mask: 11111111.11111111.11111111.00000000, which is 255.255.255.0.
- /16 means 16 ones (255.255.0.0), while /8 means 8 ones (255.0.0.0).
- Remaining host bits: subtracting the prefix from 32 gives the host bits (for /24, 32 − 24 = 8 bits, or 254 usable hosts).

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom needs addresses for 28 PCs plus a router interface.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- /24 means 24 ones in the mask: 11111111.11111111.11111111.00000000, which is 255.255.255.0.
- /16 means 16 ones (255.255.0.0), while /8 means 8 ones (255.0.0.0).
- Remaining host bits: subtracting the prefix from 32 gives the host bits (for /24, 32 − 24 = 8 bits, or 254 usable hosts).

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of cidr prefix notation (/24 shortcut).

Key Takeaway: Classless Inter-Domain Routing (CIDR) prefix notation states the number of consecutive network ones in the mask using a slash (such as /24), replacing cumbersome dotted-decimal masks.

Slide 12 of 18

Example: read one classroom address

EXAMPLE

📖 WHAT TO READ & DISPLAY ON SCREEN

An address and prefix place a host within a logical subnet. The same configuration also identifies a gateway for appropriate remote destinations.

- The /24 prefix places this host in 192.168.10.0/24. Ordinary usable host addresses run from .1 through .254.
- A peer at 192.168.10.40 is local; a destination outside this subnet needs an appropriate route, often through the configured gateway.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Understand IPv4 address structure, decimal octets, classes, private ranges, subnet boundaries, and IPv6 fundamentals.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- The /24 prefix places this host in 192.168.10.0/24.
- Ordinary usable host addresses run from .1 through .254.
- A peer at 192.168.10.40 is local; a destination outside this subnet needs an appropriate route, often through the configured gateway.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of example: read one classroom address.

Key Takeaway: An address and prefix place a host within a logical subnet. The same configuration also identifies a gateway for appropriate remote destinations.

Slide 13 of 18

Classful History vs Modern Subnetting

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Modern Classless Inter-Domain Routing (CIDR) eliminates rigid historical classes, allowing any Internet Protocol (IP) block to use any prefix size to fit the exact needs of a network.

- Classful legacy (1981): 10.0.0.0 was locked into Class A (/8), allocating 16 million hosts to one network with severe waste.
- Modern CIDR (1993): Classless Inter-Domain Routing allows any prefix to be used with any Internet Protocol (IP) address.
- 10.0.0.0/24 is normal: an administrator can carve 10.0.0.0 into small /24 subnets (254 hosts per classroom) with a 255.255.255.0 mask.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom needs addresses for 28 PCs plus a router interface.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Classful legacy (1981): 10.0.0.0 was locked into Class A (/8), allocating 16 million hosts to one network with severe waste.
- Modern CIDR (1993): Classless Inter-Domain Routing allows any prefix to be used with any IP address.
- 10.0.0.0/24 is normal: an administrator can carve 10.0.0.0 into small /24 subnets (254 hosts per classroom) with a 255.255.255.0 mask.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of classful history vs modern subnetting.

Key Takeaway: Modern Classless Inter-Domain Routing (CIDR) eliminates rigid historical classes, allowing any Internet Protocol (IP) block to use any prefix size to fit the exact needs of a network.

Slide 14 of 18

Three Private IPv4 Address Ranges

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Request for Comments (RFC) 1918 reserves three private address blocks for local networks: 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16. These addresses are free to use on local LANs and are not routed on the public Internet.

- 10.0.0.0/8: 10.0.0.0 through 10.255.255.255.
- 172.16.0.0/12: 172.16.0.0 through 172.31.255.255.
- 192.168.0.0/16: 192.168.0.0 through 192.168.255.255.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom needs addresses for 28 PCs plus a router interface.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- 10.0.0.0/8: 10.0.0.0 through 10.255.255.255.
- 172.16.0.0/12: 172.16.0.0 through 172.31.255.255.
- 192.168.0.0/16: 192.168.0.0 through 192.168.255.255.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of three private ipv4 address ranges.

Key Takeaway: Request for Comments (RFC) 1918 reserves three private address blocks for local networks: 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16. These addresses are free to use on local LANs and are not routed on the public Internet.

Slide 15 of 18

Eight Special-Purpose IPv4 Ranges

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Several address ranges have special uses such as loopback (127.0.0.1), link-local APIPA (169.254.0.0/16), and documentation. An address outside the private ranges is not automatically a globally reachable host address.

- 127.0.0.0/8: loopback.
- 169.254.0.0/16: Internet Protocol version 4 (IPv4) link-local.
- 100.64.0.0/10: shared carrier-grade Network Address Translation (NAT) space.
- 192.0.2.0/24: documentation.
- 198.51.100.0/24: documentation.
- 203.0.113.0/24: documentation.
- 224.0.0.0/4: multicast.
- 240.0.0.0/4: reserved space, including limited broadcast at 255.255.255.255.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom needs addresses for 28 PCs plus a router interface.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- 127.0.0.0/8: loopback.
- 169.254.0.0/16: IPv4 link-local.
- 100.64.0.0/10: shared carrier-grade NAT space.
- 192.0.2.0/24: documentation.
- 198.51.100.0/24: documentation.
- 203.0.113.0/24: documentation.
- 224.0.0.0/4: multicast.
- 240.0.0.0/4: reserved space, including limited broadcast at 255.255.255.255.
- This is not an exhaustive registry.
- A non-private address is not automatically ordinary globally routable unicast.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of eight special-purpose ipv4 ranges.

Key Takeaway: Several address ranges have special uses such as loopback (127.0.0.1), link-local APIPA (169.254.0.0/16), and documentation. An address outside the private ranges is not automatically a globally reachable host address.

Slide 16 of 18

Introduction to IPv6 Basics

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Internet Protocol version 6 (IPv6) solves Internet Protocol version 4 (IPv4) address exhaustion by providing 128-bit addresses, offering practically unlimited addresses for modern global devices.

- Why Internet Protocol version 6 (IPv6) is needed: Internet Protocol version 4 (IPv4) provides only 4.3 billion addresses, which are now completely exhausted by billions of smartphones, computers, and cloud servers.
- 128-Bit Structure: Internet Protocol version 6 (IPv6) uses 128 binary bits written as eight groups of four hexadecimal digits separated by colons (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).
- Zero Compression Rule (::): Consecutive groups of zeros can be abbreviated with a double colon (::) once in an address (e.g., 2001:db8::1).
- Address Types Without Broadcast: Internet Protocol version 6 (IPv6) uses Unicast (one-to-one), Multicast (one-to-many), and Anycast (one-to-nearest). IPv6 eliminates noisy broadcast messages entirely.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom needs addresses for 28 PCs plus a router interface.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Why IPv6 is needed: IPv4 provides only 4.3 billion addresses, which are now completely exhausted by billions of smartphones, computers, and cloud servers.
- 128-Bit Structure: IPv6 uses 128 binary bits written as eight groups of four hexadecimal digits separated by colons (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).
- Zero Compression Rule (::): Consecutive groups of zeros can be abbreviated with a double colon (::) once in an address (e.g., 2001:db8::1).
- Address Types Without Broadcast: IPv6 uses Unicast (one-to-one), Multicast (one-to-many), and Anycast (one-to-nearest).
- IPv6 eliminates noisy broadcast messages entirely.
- IPv6 provides 340 undecillion addresses (3.4 × 10^38), ensuring every device on Earth can have its own permanent, globally unique IP address without needing NAT.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of introduction to ipv6 basics.

Key Takeaway: Internet Protocol version 6 (IPv6) solves Internet Protocol version 4 (IPv4) address exhaustion by providing 128-bit addresses, offering practically unlimited addresses for modern global devices.

Slide 17 of 18

Subnet Sizing and Address Planning

APPLICATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Determine the required host capacity before selecting a subnet prefix. Sizing must account for gateway routers, reserved addresses, and planned future growth.

- /28 offers only 14 ordinary usable addresses, which is insufficient.
- /27 gives 30 usable addresses: 28 PCs, 1 router, 1 spare.
- Allocate addresses sequentially and document the plan.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom needs addresses for 28 PCs plus a router interface.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- /28 offers only 14 ordinary usable addresses, which is insufficient.
- /27 gives 30 usable addresses: 28 PCs, 1 router, 1 spare.
- Allocate addresses sequentially and document the plan.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of subnet sizing and address planning.

Key Takeaway: Determine the required host capacity before selecting a subnet prefix. Sizing must account for gateway routers, reserved addresses, and planned future growth.

TOPIC 8 OF 13

Topic 8: Static Internet Protocol (IP) Configuration

Configure static IP addresses on Windows hosts, verify local vs. remote delivery, and test network reachability.

Curriculum: Networking 1

Excluded: Title (Slide 1), Objectives (Slide 2), and Summary (Final Slide)

Discussion Units: 11 Slides

Slide 3 of 14

Subnet Mask, Gateway, and DNS Roles

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

The subnet mask defines the local network boundary, the default gateway forwards remote traffic to the outside world, and Domain Name System (DNS) resolves hostnames to numeric Internet Protocol (IP) addresses.

- 192.168.1.10/24 uses 24 network bits and 8 host bits; its mask is 255.255.255.0.
- The default gateway is a local router used for destinations without a more specific route.
- Domain Name System (DNS): maps names to records such as Internet Protocol (IP) addresses.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Inspect host settings and assign a stable address to a shared network printer.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- 192.168.1.10/24 uses 24 network bits and 8 host bits; its mask is 255.255.255.0.
- The default gateway is a local router used for destinations without a more specific route.
- DNS: maps names to records such as IP addresses.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of subnet mask, gateway, and dns roles.

Key Takeaway: The subnet mask defines the local network boundary, the default gateway forwards remote traffic to the outside world, and Domain Name System (DNS) resolves hostnames to numeric Internet Protocol (IP) addresses.

Slide 4 of 14

Two Ways to Assign an IP Address

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

An address can be configured manually or obtained through a configuration service. Static and dynamic assignment differ in how settings are supplied and maintained.

- Static: an administrator manually configures the address and related settings.
- Dynamic: Dynamic Host Configuration Protocol (DHCP) supplies an address lease and configuration options.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Inspect host settings and assign a stable address to a shared network printer.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Static: an administrator manually configures the address and related settings.
- Dynamic: DHCP supplies an address lease and configuration options.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of two ways to assign an ip address.

Key Takeaway: An address can be configured manually or obtained through a configuration service. Static and dynamic assignment differ in how settings are supplied and maintained.

Slide 5 of 14

What Is Static IP Configuration?

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Static configuration assigns an interface's address settings manually. A predictable address helps infrastructure and services remain locatable when settings are managed correctly.

- Record address, mask, gateway, and Domain Name System (DNS) before changing anything. A Dynamic Host Configuration Protocol (DHCP) reservation is a different way to provide a stable address.
- Choose a unique usable address in the intended subnet and coordinate it with the administrator's addressing plan.
- Avoid unmanaged overlap with Dynamic Host Configuration Protocol (DHCP) pools. Duplicate addresses cause conflicts even when each individual address is syntactically valid.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Inspect host settings and assign a stable address to a shared network printer.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Record address, mask, gateway, and DNS before changing anything.
- A DHCP reservation is a different way to provide a stable address.
- Choose a unique usable address in the intended subnet and coordinate it with the administrator's addressing plan.
- Avoid unmanaged overlap with DHCP pools.
- Duplicate addresses cause conflicts even when each individual address is syntactically valid.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of what is static ip configuration?.

Key Takeaway: Static configuration assigns an interface's address settings manually. A predictable address helps infrastructure and services remain locatable when settings are managed correctly.

Slide 6 of 14

Four Key IPv4 Configuration Settings

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A basic host configuration includes an address, subnet mask, default gateway, and name resolver. Each field has a separate purpose and must match the intended network.

- Internet Protocol (IP) address: a unique usable address assigned to the interface.
- Subnet mask: identifies which address bits describe the local network.
- Default gateway: a local router for destinations needing a remote route.
- Domain Name System (DNS) server: a resolver for names; not needed for numeric-IP local tests.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Inspect host settings and assign a stable address to a shared network printer.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- IP address: a unique usable address assigned to the interface.
- Subnet mask: identifies which address bits describe the local network.
- Default gateway: a local router for destinations needing a remote route.
- DNS server: a resolver for names; not needed for numeric-IP local tests.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of four key ipv4 configuration settings.

Key Takeaway: A basic host configuration includes an address, subnet mask, default gateway, and name resolver. Each field has a separate purpose and must match the intended network.

Slide 7 of 14

How Hosts Decide if a Peer is Local

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Each host uses its own address and mask to decide whether a peer is local. A successful local round trip needs compatible decisions and a working link in both directions.

- They can communicate by Internet Protocol (IP) on a correctly connected Local Area Network (LAN) without a default gateway.
- 192.168.1.10/24 and 192.168.2.20/24 need appropriate routing between the two subnets.
- Mismatched masks can make one host consider a destination local while the other sends through a gateway. Test both directions.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Inspect host settings and assign a stable address to a shared network printer.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- They can communicate by IP on a correctly connected LAN without a default gateway.
- 192.168.1.10/24 and 192.168.2.20/24 need appropriate routing between the two subnets.
- Mismatched masks can make one host consider a destination local while the other sends through a gateway.
- Test both directions.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of how hosts decide if a peer is local.

Key Takeaway: Each host uses its own address and mask to decide whether a peer is local. A successful local round trip needs compatible decisions and a working link in both directions.

Slide 8 of 14

Example: two manually configured classroom computers

EXAMPLE

📖 WHAT TO READ & DISPLAY ON SCREEN

Two hosts with unique usable addresses in the same subnet can communicate over a working local network. A default gateway is unnecessary for this numeric local exchange.

- Both hosts calculate 192.168.1.0/24 as their local network, so this exchange does not require a router.
- The addresses must be unique and the link and host policy must permit communication. A successful address calculation alone does not prove reachability.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Configure static IP addresses on Windows hosts, verify local vs. remote delivery, and test network reachability.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Both hosts calculate 192.168.1.0/24 as their local network, so this exchange does not require a router.
- The addresses must be unique and the link and host policy must permit communication.
- A successful address calculation alone does not prove reachability.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of example: two manually configured classroom computers.

Key Takeaway: Two hosts with unique usable addresses in the same subnet can communicate over a working local network. A default gateway is unnecessary for this numeric local exchange.

Slide 9 of 14

When Subnet Masks Do Not Match

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Different subnet masks can make two hosts disagree about whether they share a local network. One host may try direct delivery while the other requires a route that is unavailable.

Interactive Demonstration: `asymmetricDemo` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Inspect host settings and assign a stable address to a shared network printer.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Calculate each host's local/remote decision independently using its own mask.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of when subnet masks do not match.

Key Takeaway: Different subnet masks can make two hosts disagree about whether they share a local network. One host may try direct delivery while the other requires a route that is unavailable.

Slide 10 of 14

Configuring Static IP in Windows

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Windows adapter settings control an interface's address and related options. Configuration must be verified through both inspection and an appropriate connectivity test.

- Open Settings → Network & Internet → Advanced network settings → More network adapter options (or run ncpa.cpl).
- Open the intended adapter's Properties → Internet Protocol Version 4 (Transmission Control Protocol (TCP)/IPv4) → Properties. Enter the approved values, then confirm.
- Use ipconfig /all and ping the peer's numeric address. Record results; restore "Obtain an Internet Protocol (IP) address automatically" and automatic Domain Name System (DNS) if that was the original setup.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Inspect host settings and assign a stable address to a shared network printer.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Open Settings → Network & Internet → Advanced network settings → More network adapter options (or run ncpa.cpl).
- Open the intended adapter's Properties → Internet Protocol Version 4 (TCP/IPv4) → Properties.
- Enter the approved values, then confirm.
- Use ipconfig /all and ping the peer's numeric address.
- Record results; restore "Obtain an IP address automatically" and automatic DNS if that was the original setup.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of configuring static ip in windows.

Key Takeaway: Windows adapter settings control an interface's address and related options. Configuration must be verified through both inspection and an appropriate connectivity test.

Slide 11 of 14

Windows Static Address Configuration Procedure

WALKTHROUGH

📖 WHAT TO READ & DISPLAY ON SCREEN

Manual address configuration requires the intended adapter and a coordinated set of valid settings. Recording the original configuration supports verification and restoration after the change.

- Record the original configuration and select the intended adapter. Open ncpa.cpl, adapter Properties, then Internet Protocol Version 4 (Transmission Control Protocol (TCP)/IPv4) Properties.
- On an approved isolated Local Area Network (LAN), 192.168.1.10/24 and 192.168.1.20/24 are compatible local hosts. A gateway is unnecessary for a numeric local test.
- Use ipconfig /all to inspect settings, then test the peer in both directions. If Internet Control Message Protocol (ICMP) is filtered, examine approved policy rather than disabling the firewall.
- Changing the second host to 192.168.2.20/24 requires appropriate gateways and bidirectional routing for cross-subnet communication.
- Restore the original approved Internet Protocol (IP)/Domain Name System (DNS) method after a physical demonstration; browser settings do not change the real Windows adapter.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Inspect host settings and assign a stable address to a shared network printer.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Record the original configuration and select the intended adapter.
- Open ncpa.cpl, adapter Properties, then Internet Protocol Version 4 (TCP/IPv4) Properties.
- On an approved isolated LAN, 192.168.1.10/24 and 192.168.1.20/24 are compatible local hosts.
- A gateway is unnecessary for a numeric local test.
- Use ipconfig /all to inspect settings, then test the peer in both directions.
- If ICMP is filtered, examine approved policy rather than disabling the firewall.
- Changing the second host to 192.168.2.20/24 requires appropriate gateways and bidirectional routing for cross-subnet communication.
- Restore the original approved IP/DNS method after a physical demonstration; browser settings do not change the real Windows adapter.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of windows static address configuration procedure.

Key Takeaway: Manual address configuration requires the intended adapter and a coordinated set of valid settings. Recording the original configuration supports verification and restoration after the change.

Slide 12 of 14

Static IP Compatibility and Testing

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Syntactically valid addresses do not guarantee that two hosts can communicate. Local subnet agreement, link availability, and any required routes determine the modeled path.

Interactive Demonstration: `static` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Inspect host settings and assign a stable address to a shared network printer.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Modeled network • validate fields, disconnect the link, or enable a router with interfaces in both subnets.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of static ip compatibility and testing.

Key Takeaway: Syntactically valid addresses do not guarantee that two hosts can communicate. Local subnet agreement, link availability, and any required routes determine the modeled path.

Slide 13 of 14

Configuring a Static LAN Printer

APPLICATION

📖 WHAT TO READ & DISPLAY ON SCREEN

A classroom host can use the same router address for its gateway and forwarded name-resolution service, while a printer benefits from an address coordinated outside the dynamic Dynamic Host Configuration Protocol (DHCP) pool.

- The host and gateway share a local subnet; the router may forward Domain Name System (DNS) requests.
- The printer benefits from an address excluded from the Dynamic Host Configuration Protocol (DHCP) dynamic pool to prevent conflicts.
- Verify numeric connectivity and service ports separately; a successful ping does not prove application reachability.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Inspect host settings and assign a stable address to a shared network printer.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- The host and gateway share a local subnet; the router may forward DNS requests.
- The printer benefits from an address excluded from the DHCP dynamic pool to prevent conflicts.
- Verify numeric connectivity and service ports separately; a successful ping does not prove application reachability.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of configuring a static lan printer.

Key Takeaway: A classroom host can use the same router address for its gateway and forwarded name-resolution service, while a printer benefits from an address coordinated outside the dynamic Dynamic Host Configuration Protocol (DHCP) pool.

TOPIC 9 OF 13

Topic 9: Network Protocols and Services

Follow common network services, compare their functions and security, and inspect transport ports through controlled demonstrations.

Curriculum: Networking 1

Excluded: Title (Slide 1), Objectives (Slide 2), and Summary (Final Slide)

Discussion Units: 19 Slides

Slide 3 of 22

Three Core Terms: Protocol, Service, Port

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A protocol defines exchange rules, a service provides a useful function, and a transport port acts like an apartment door number directing traffic to a specific application process.

- Protocol: agreed message formats and behavior, such as Domain Name System (DNS) queries and replies.
- Service: a useful function provided by a running system, such as a Domain Name System (DNS) resolver answering names.
- Port: a Transmission Control Protocol (TCP) or User Datagram Protocol (UDP) endpoint number used to direct traffic to a process; it is not a physical Ethernet socket.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Protocol: agreed message formats and behavior, such as DNS queries and replies.
- Service: a useful function provided by a running system, such as a DNS resolver answering names.
- Port: a TCP or UDP endpoint number used to direct traffic to a process; it is not a physical Ethernet socket.
- A familiar port suggests a convention; it does not prove which program is listening or whether that program is safe.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of three core terms: protocol, service, port.

Key Takeaway: A protocol defines exchange rules, a service provides a useful function, and a transport port acts like an apartment door number directing traffic to a specific application process.

Slide 4 of 22

Four TCP/IP responsibility groups

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

The common Transmission Control Protocol (TCP)/Internet Protocol (IP) model groups network functions into four responsibility areas. Application services depend on transport, Internet delivery, and local-link functions.

- Application: Dynamic Host Configuration Protocol (DHCP), Domain Name System (DNS), Hypertext Transfer Protocol (HTTP), Hypertext Transfer Protocol Secure (HTTPS), and Secure Shell (SSH) provide user-facing and network configuration services; commonly associated with Open Systems Interconnection (OSI) 7.
- Transport: Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) use process ports; commonly associated with Open Systems Interconnection (OSI) 4.
- Internet: Internet Protocol (IP) supplies logical delivery and Internet Control Message Protocol (ICMP) supplies related diagnostics; commonly associated with Open Systems Interconnection (OSI) 3.
- Link: local framing and physical transmission; Address Resolution Protocol (ARP) supports Internet Protocol version 4 (IPv4) neighbor resolution at this local-link boundary.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Application: DHCP, DNS, HTTP, HTTPS, and SSH provide user-facing and network configuration services; commonly associated with OSI 7.
- Transport: TCP and UDP use process ports; commonly associated with OSI 4.
- Internet: IP supplies logical delivery and ICMP supplies related diagnostics; commonly associated with OSI 3.
- Link: local framing and physical transmission; ARP supports IPv4 neighbor resolution at this local-link boundary.
- OSI/TCP-IP mappings are approximate.
- Advanced enterprise protocols like SNMP, NTP, and dedicated mail servers are studied in Networking II.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of four tcp/ip responsibility groups.

Key Takeaway: The common Transmission Control Protocol (TCP)/Internet Protocol (IP) model groups network functions into four responsibility areas. Application services depend on transport, Internet delivery, and local-link functions.

Slide 5 of 22

Two transport services: TCP and UDP

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Transmission Control Protocol supplies ordered reliable bytes, while User Datagram Protocol carries independent datagrams without built-in delivery guarantees. Their differences affect how application protocols handle communication.

- Transmission Control Protocol (TCP): connection-oriented, ordered reliable bytes, with acknowledgments and retransmission. Useful when the application needs this stream service.
- User Datagram Protocol (UDP): connectionless datagrams without built-in delivery or ordering guarantees. Useful when a small exchange or application-controlled behavior fits the task.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- TCP: connection-oriented, ordered reliable bytes, with acknowledgments and retransmission.
- Useful when the application needs this stream service.
- UDP: connectionless datagrams without built-in delivery or ordering guarantees.
- Useful when a small exchange or application-controlled behavior fits the task.
- Neither “UDP is always faster” nor “TCP proves the application processed the request” is a reliable conclusion.
- Application protocols can add recovery above UDP.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of two transport services: tcp and udp.

Key Takeaway: Transmission Control Protocol supplies ordered reliable bytes, while User Datagram Protocol carries independent datagrams without built-in delivery guarantees. Their differences affect how application protocols handle communication.

Slide 6 of 22

Client and Server Endpoints

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A client commonly uses a temporary source port to contact a server's listening port. Addresses, ports, and transport together identify the communication context.

- A normal client often uses a temporary source port, such as Transmission Control Protocol (TCP) 53001, to contact a server listening on TCP 443. The response returns to that client endpoint.
- The connection context includes transport plus source/destination addresses and ports. Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) can use the same numeric port independently.
- System ports are 0–1023, user/registered ports 1024–49151, and dynamic/private ports 49152–65535. Port 0 is reserved; operating systems may choose temporary ports from implementation-specific ranges.
- A reachable Internet Protocol (IP), an open transport endpoint, Transport Layer Security (TLS) validation, and a working application are separate observations. Testing one does not prove all the others.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- A normal client often uses a temporary source port, such as TCP 53001, to contact a server listening on TCP 443.
- The response returns to that client endpoint.
- The connection context includes transport plus source/destination addresses and ports.
- TCP and UDP can use the same numeric port independently.
- System ports are 0–1023, user/registered ports 1024–49151, and dynamic/private ports 49152–65535.
- Port 0 is reserved; operating systems may choose temporary ports from implementation-specific ranges.
- A reachable IP, an open transport endpoint, TLS validation, and a working application are separate observations.
- Testing one does not prove all the others.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of client and server endpoints.

Key Takeaway: A client commonly uses a temporary source port to contact a server's listening port. Addresses, ports, and transport together identify the communication context.

Slide 7 of 22

Transport Reliability: TCP vs UDP

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Transport protocols differ in their response to missing data. Transmission Control Protocol provides recovery and ordered bytes, while User Datagram Protocol leaves any needed recovery to the application.

Interactive Demonstration: `osiLayer` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Compare TCP and UDP using the same modeled missing data unit.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of transport reliability: tcp vs udp.

Key Takeaway: Transport protocols differ in their response to missing data. Transmission Control Protocol provides recovery and ordered bytes, while User Datagram Protocol leaves any needed recovery to the application.

Slide 8 of 22

Internet Protocol (IP)

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Internet Protocol supplies logical addressing and packet delivery across networks. It provides best-effort delivery rather than an ordered reliable application stream.

- Practical application: Internet Protocol version 4 (IPv4) or Internet Protocol version 6 (IPv6) identifies source/destination interfaces and supports router forwarding.
- Internet Protocol (IP) provides best-effort delivery, not a reliable application byte stream. It has no Transmission Control Protocol (TCP)/User Datagram Protocol (UDP) port. Internet Protocol version 6 (IPv6) uses 128-bit addresses and does not use Address Resolution Protocol (ARP).

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Practical application: IPv4 or IPv6 identifies source/destination interfaces and supports router forwarding.
- IP provides best-effort delivery, not a reliable application byte stream.
- It has no TCP/UDP port.
- IPv6 uses 128-bit addresses and does not use ARP.
- OSI 3 · TCP/IP Internet.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of internet protocol (ip).

Key Takeaway: Internet Protocol supplies logical addressing and packet delivery across networks. It provides best-effort delivery rather than an ordered reliable application stream.

Slide 9 of 22

Address Resolution Protocol (ARP)

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Address Resolution Protocol resolves the hardware address of an Internet Protocol version 4 (IPv4) next hop on a local link. A remote destination usually requires the gateway's local hardware address rather than the remote host's.

- Practical application: Find a local peer's Ethernet Media Access Control (MAC), or the gateway MAC when sending toward a remote Internet Protocol (IP) network.
- Address Resolution Protocol (ARP) is not carried inside Transmission Control Protocol (TCP) or User Datagram Protocol (UDP) and has no port. Ethernet ARP uses EtherType 0x0806. Routers do not forward a local ARP broadcast across Internet Protocol (IP) networks; Internet Protocol version 6 (IPv6) uses Neighbor Discovery instead.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Practical application: Find a local peer's Ethernet MAC, or the gateway MAC when sending toward a remote IP network.
- ARP is not carried inside TCP or UDP and has no port.
- Ethernet ARP uses EtherType 0x0806.
- Routers do not forward a local ARP broadcast across IP networks; IPv6 uses Neighbor Discovery instead.
- Local-link boundary · TCP/IP Link; commonly associated with OSI 2.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of address resolution protocol (arp).

Key Takeaway: Address Resolution Protocol resolves the hardware address of an Internet Protocol version 4 (IPv4) next hop on a local link. A remote destination usually requires the gateway's local hardware address rather than the remote host's.

Slide 10 of 22

How ARP Resolves MAC Addresses

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

A host needs a local delivery address before sending an Ethernet frame toward an Internet Protocol version 4 (IPv4) neighbor. Address Resolution Protocol uses a local request and reply to establish that mapping.

Interactive Demonstration: `protocolArp` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Change between a local peer and a remote destination; follow the calculated next-hop explanation.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of how arp resolves mac addresses.

Key Takeaway: A host needs a local delivery address before sending an Ethernet frame toward an Internet Protocol version 4 (IPv4) neighbor. Address Resolution Protocol uses a local request and reply to establish that mapping.

Slide 11 of 22

ICMP and Network Ping Basics

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Internet Control Message Protocol carries diagnostic and error information related to Internet Protocol delivery. Echo messages and expiry reports provide evidence without proving that every application service works.

- Practical application: Ping commonly uses Echo; traceroute-style tools use Time to Live (TTL)/hop-limit behavior and returned diagnostics.
- Internet Control Message Protocol (ICMP) has no Transmission Control Protocol (TCP)/User Datagram Protocol (UDP) port. ICMP for Internet Protocol version 4 (IPv4) uses Internet Protocol (IP) protocol number 1; Internet Control Message Protocol for IPv6 (ICMPv6) uses next-header 58. Missing replies may mean filtering or loss. It does not fix a failed link.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Practical application: Ping commonly uses Echo; traceroute-style tools use TTL/hop-limit behavior and returned diagnostics.
- ICMP has no TCP/UDP port.
- ICMP for IPv4 uses IP protocol number 1; ICMPv6 uses next-header 58.
- Missing replies may mean filtering or loss.
- It does not fix a failed link.
- OSI 3 · TCP/IP Internet.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of icmp and network ping basics.

Key Takeaway: Internet Control Message Protocol carries diagnostic and error information related to Internet Protocol delivery. Echo messages and expiry reports provide evidence without proving that every application service works.

Slide 12 of 22

Dynamic Host Configuration Protocol (DHCP)

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Dynamic Host Configuration Protocol supplies leased host configuration and related options. Clients use it to obtain settings such as address, mask, gateway, and name resolver.

- Practical application: Automatically configure classroom PCs without entering each address manually.
- Initial allocation is explained as Discover, Offer, Request, Acknowledgment (ACK). The ports below describe DHCPv4; DHCPv6 has different messages and ports.
- User Datagram Protocol (UDP) 67: DHCPv4 server.
- User Datagram Protocol (UDP) 68: DHCPv4 client.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Practical application: Automatically configure classroom PCs without entering each address manually.
- Initial allocation is explained as Discover, Offer, Request, ACK.
- The ports below describe DHCPv4; DHCPv6 has different messages and ports.
- UDP 67: DHCPv4 server.
- UDP 68: DHCPv4 client.
- OSI 7 · TCP/IP Application.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of dynamic host configuration protocol (dhcp).

Key Takeaway: Dynamic Host Configuration Protocol supplies leased host configuration and related options. Clients use it to obtain settings such as address, mask, gateway, and name resolver.

Slide 13 of 22

DHCP Ports and Communication

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Dynamic Host Configuration Protocol version 4 uses server User Datagram Protocol port 67 and client port 68. Its initial exchange proposes, requests, and confirms an address configuration.

Interactive Demonstration: dhcp — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- DHCPv4 uses server UDP 67 and client UDP 68; a lease becomes active at ACK.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of dhcp ports and communication.

Key Takeaway: Dynamic Host Configuration Protocol version 4 uses server User Datagram Protocol port 67 and client port 68. Its initial exchange proposes, requests, and confirms an address configuration.

Slide 14 of 22

Domain Name System (DNS)

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

The Domain Name System answers queries about names using records such as host addresses and mail destinations. Name resolution supplies information for later communication rather than carrying the application traffic itself.

- Practical application: Find the address for a school web or mail server by name.
- Classic Domain Name System (DNS) supports User Datagram Protocol (UDP) and Transmission Control Protocol (TCP) on port 53. TCP is not limited to zone transfers: it also supports other queries, including fallback after truncation. Caches respect record lifetimes. Encrypted DNS has additional transports.
- User Datagram Protocol (UDP) 53: Classic Domain Name System (DNS) queries/responses.
- Transmission Control Protocol (TCP) 53: Classic Domain Name System (DNS) over a TCP connection.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Practical application: Find the address for a school web or mail server by name.
- Classic DNS supports UDP and TCP on port 53.
- TCP is not limited to zone transfers: it also supports other queries, including fallback after truncation.
- Caches respect record lifetimes.
- Encrypted DNS has additional transports.
- UDP 53: Classic DNS queries/responses.
- TCP 53: Classic DNS over a TCP connection.
- OSI 7 · TCP/IP Application.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of domain name system (dns).

Key Takeaway: The Domain Name System answers queries about names using records such as host addresses and mail destinations. Name resolution supplies information for later communication rather than carrying the application traffic itself.

Slide 15 of 22

DNS Caching and Record Lifetimes

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

A cached name answer can be reused until its permitted lifetime expires. Server record changes may therefore remain unseen by a client until its cache entry is refreshed.

Interactive Demonstration: `dnsCacheDemo` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Follow common network services, compare their functions and security, and inspect transport ports through controlled demonstrations.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Resolve names, change a server record, and advance the client cache lifetime.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of dns caching and record lifetimes.

Key Takeaway: A cached name answer can be reused until its permitted lifetime expires. Server record changes may therefore remain unseen by a client until its cache entry is refreshed.

Slide 16 of 22

Hypertext Transfer Protocol (HTTP)

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Hypertext Transfer Protocol defines requests and responses for web resources. Methods, status codes, and message content describe the requested operation and its result.

- Practical application: A browser requests a page or a client calls a web Application Programming Interface (API).
- Ordinary Hypertext Transfer Protocol (HTTP) does not provide Transport Layer Security (TLS) encryption. HTTP/1.1 commonly uses Transmission Control Protocol (TCP); port 80 is the conventional HTTP service endpoint. An HTTP status error differs from a failure to reach the server.
- Transmission Control Protocol (TCP) 80: Conventional unencrypted Hypertext Transfer Protocol (HTTP) service.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Practical application: A browser requests a page or a client calls a web API.
- Ordinary HTTP does not provide TLS encryption.
- HTTP/1.1 commonly uses TCP; port 80 is the conventional HTTP service endpoint.
- An HTTP status error differs from a failure to reach the server.
- TCP 80: Conventional unencrypted HTTP service.
- OSI 7 · TCP/IP Application.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of hypertext transfer protocol (http).

Key Takeaway: Hypertext Transfer Protocol defines requests and responses for web resources. Methods, status codes, and message content describe the requested operation and its result.

Slide 17 of 22

HTTPS and Secure Web Browsing

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Hypertext Transfer Protocol Secure protects web communication using Transport Layer Security. Certificate validation and authenticated encryption protect the connection without guaranteeing that the website's content is trustworthy.

- Practical application: Access a protected classroom portal without exposing application content on the local path.
- Hypertext Transfer Protocol (HTTP)/1.1 and HTTP/2 commonly use Transport Layer Security (TLS) over Transmission Control Protocol (TCP) 443. HTTP/3 uses QUIC with TLS security over User Datagram Protocol (UDP), commonly 443. Hypertext Transfer Protocol Secure (HTTPS) does not prove a site is trustworthy or hide every piece of network metadata.
- Transmission Control Protocol (TCP) 443: Hypertext Transfer Protocol Secure (HTTPS) using Hypertext Transfer Protocol (HTTP)/1.1 or HTTP/2.
- User Datagram Protocol (UDP) 443: Hypertext Transfer Protocol (HTTP)/3 over QUIC; common Hypertext Transfer Protocol Secure (HTTPS) endpoint.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Practical application: Access a protected classroom portal without exposing application content on the local path.
- HTTP/1.1 and HTTP/2 commonly use TLS over TCP 443.
- HTTP/3 uses QUIC with TLS security over UDP, commonly 443.
- HTTPS does not prove a site is trustworthy or hide every piece of network metadata.
- TCP 443: HTTPS using HTTP/1.1 or HTTP/2.
- UDP 443: HTTP/3 over QUIC; common HTTPS endpoint.
- OSI 7 service · TCP/IP Application; TLS spans security/representation responsibilities.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of https and secure web browsing.

Key Takeaway: Hypertext Transfer Protocol Secure protects web communication using Transport Layer Security. Certificate validation and authenticated encryption protect the connection without guaranteeing that the website's content is trustworthy.

Slide 18 of 22

Example: open a protected school website

EXAMPLE

📖 WHAT TO READ & DISPLAY ON SCREEN

Opening a protected website combines name resolution with a secure application exchange. The resolver supplies an address, while the browser exchanges web traffic with the server.

- Domain Name System (DNS) resolves the server name. For a common Hypertext Transfer Protocol (HTTP)/1.1-over-TLS example, Transmission Control Protocol (TCP) connects to server port 443 and Transport Layer Security (TLS) protects the exchange.
- Hypertext Transfer Protocol Secure (HTTPS) carries the web request and response inside that protected connection. The resolver is not a relay for the web page.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Follow common network services, compare their functions and security, and inspect transport ports through controlled demonstrations.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Domain Name System (DNS) resolves the server name.
- For a common HTTP/1.1-over-TLS example, Transmission Control Protocol (TCP) connects to server port 443 and Transport Layer Security (TLS) protects the exchange.
- Hypertext Transfer Protocol Secure (HTTPS) carries the web request and response inside that protected connection.
- The resolver is not a relay for the web page.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of example: open a protected school website.

Key Takeaway: Opening a protected website combines name resolution with a secure application exchange. The resolver supplies an address, while the browser exchanges web traffic with the server.

Slide 19 of 22

How Protocols Cooperate on the Web

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

A web request depends on local delivery, name resolution, transport, connection security, and the web service. A failure at one stage produces different evidence from a failure at another.

Interactive Demonstration: `protocolWeb` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Follow ARP, classic DNS, IP, TCP, TLS, and HTTP; change a condition to see exactly where the operation differs.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of how protocols cooperate on the web.

Key Takeaway: A web request depends on local delivery, name resolution, transport, connection security, and the web service. A failure at one stage produces different evidence from a failure at another.

Slide 20 of 22

Secure Shell (SSH)

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Secure Shell provides authenticated encrypted remote access and channels. Checking server identity and user authorization is part of establishing a trusted connection.

- Practical application: Administer an authorized router/server or carry an encrypted administrative session.
- Check the server host identity and authenticate using an authorized account. A port number alone does not establish that authentication or encryption is correctly configured.
- Transmission Control Protocol (TCP) 22: Conventional Secure Shell (SSH) server endpoint.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Practical application: Administer an authorized router/server or carry an encrypted administrative session.
- Check the server host identity and authenticate using an authorized account.
- A port number alone does not establish that authentication or encryption is correctly configured.
- TCP 22: Conventional SSH server endpoint.
- OSI 7 · TCP/IP Application.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of secure shell (ssh).

Key Takeaway: Secure Shell provides authenticated encrypted remote access and channels. Checking server identity and user authorization is part of establishing a trusted connection.

Slide 21 of 22

Common Protocols and Port Numbers

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Protocol and port references identify conventional service roles and endpoint bindings. The transport must accompany the number, and some network protocols have no transport port at all.

Interactive Demonstration: `protocolDirectory` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Inspect all ten core protocols individually, including transport and layer relationships.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of common protocols and port numbers.

Key Takeaway: Protocol and port references identify conventional service roles and endpoint bindings. The transport must accompany the number, and some network protocols have no transport port at all.

TOPIC 10 OF 13

Topic 10: Dynamic Host Configuration Protocol (DHCP)

Step through DORA and manage a real simulated pool with leases, renewal, and exhaustion.

Curriculum: Networking 1 **Excluded:** Title (Slide 1), Objectives (Slide 2), and Summary (Final Slide) **Discussion Units:** 8 Slides

Slide 3 of 11

What Is a DHCP Lease?

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A lease grants temporary use of an address supplied by Dynamic Host Configuration Protocol. The server also provides options such as mask, gateway, and name resolver.

- A pool is the set of available addresses. Exclusions and reservations keep infrastructure addresses under control.
- A lease grants temporary use; Dynamic Host Configuration Protocol (DHCP) also supplies options such as subnet mask, gateway, and Domain Name System (DNS).
- A server must avoid duplicate active assignments. Clients and servers also use conflict detection in real deployments.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *The router has a DHCP pool of only twenty addresses and several existing leases.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- A pool is the set of available addresses.
- Exclusions and reservations keep infrastructure addresses under control.
- A lease grants temporary use; DHCP also supplies options such as subnet mask, gateway, and DNS.
- A server must avoid duplicate active assignments.
- Clients and servers also use conflict detection in real deployments.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of what is a dhcp lease?.

Key Takeaway: A lease grants temporary use of an address supplied by Dynamic Host Configuration Protocol. The server also provides options such as mask, gateway, and name resolver.

Slide 4 of 11

Four messages in the initial DHCP exchange

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Initial allocation commonly follows Discover, Offer, Request, and Acknowledgment. These messages locate a server, propose settings, select an offer, and confirm the lease.

- Discover: the client seeks Dynamic Host Configuration Protocol (DHCP) servers.
- Offer: a server proposes an available address and configuration options.
- Request: the client identifies the chosen offer and requests its configuration.
- Acknowledgment (ACK): the server confirms the lease, after which the client can use the configuration.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *The router has a DHCP pool of only twenty addresses and several existing leases.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Discover: the client seeks DHCP servers.
- Offer: a server proposes an available address and configuration options.
- Request: the client identifies the chosen offer and requests its configuration.
- Acknowledgment (ACK): the server confirms the lease, after which the client can use the configuration.
- Initial discovery uses broadcast.
- Some replies can be unicast depending on client state; DORA is an introduction to a richer state machine.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of four messages in the initial dhcp exchange.

Key Takeaway: Initial allocation commonly follows Discover, Offer, Request, and Acknowledgment. These messages locate a server, propose settings, select an offer, and confirm the lease.

Slide 5 of 11

Example: a laptop receives a lease

EXAMPLE

📖 WHAT TO READ & DISPLAY ON SCREEN

A new laptop can obtain address settings from an available configuration server. The confirmed lease must use an address that does not conflict with another active allocation.

- The laptop discovers the service, receives an offer for 192.168.1.100, requests it, and receives an acknowledgment committing the lease.
- The server also supplies subnet mask, gateway, resolver information, and lease lifetime. Another active client must receive a different available address.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Step through DORA and manage a real simulated pool with leases, renewal, and exhaustion.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- The laptop discovers the service, receives an offer for 192.168.1.100, requests it, and receives an acknowledgment committing the lease.
- The server also supplies subnet mask, gateway, resolver information, and lease lifetime.
- Another active client must receive a different available address.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of example: a laptop receives a lease.

Key Takeaway: A new laptop can obtain address settings from an available configuration server. The confirmed lease must use an address that does not conflict with another active allocation.

Slide 6 of 11

Five lease lifecycle and failure events

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Leases can be renewed, released, or allowed to expire. Unavailable servers and exhausted pools create different failures even when the client's physical connection works.

- Renewal: requests an extension from the existing server; it need not repeat the initial full Discover, Offer, Request, Acknowledge (DORA) exchange.
- Release: returns the address to the server.
- Expiry: makes an unrenewed lease available again.
- Pool exhaustion: no unused address is available for another client.
- Server unavailable: a client may eventually self-assign link-local configuration without ordinary routed access.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *The router has a DHCP pool of only twenty addresses and several existing leases.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Renewal: requests an extension from the existing server; it need not repeat the initial full DORA exchange.
- Release: returns the address to the server.
- Expiry: makes an unrenewed lease available again.
- Pool exhaustion: no unused address is available for another client.
- Server unavailable: a client may eventually self-assign link-local configuration without ordinary routed access.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of five lease lifecycle and failure events.

Key Takeaway: Leases can be renewed, released, or allowed to expire. Unavailable servers and exhausted pools create different failures even when the client's physical connection works.

Slide 7 of 11

Reading a DHCP Lease Configuration

WALKTHROUGH

📖 WHAT TO READ & DISPLAY ON SCREEN

Automatic configuration includes more than a host address. Lease lifetime, gateway, resolver, and server information help explain which settings the client received.

- An authorized router pool contains usable addresses inside its Local Area Network (LAN) subnet, excluding the router and other managed infrastructure.
- A client set to automatic Internet Protocol (IP)/Domain Name System (DNS) obtains address, mask, gateway, DNS, and a lease lifetime from the reachable server.
- Discover locates a server, Offer proposes configuration, Request selects it, and Acknowledgment (ACK) commits the lease.
- On authorized Windows equipment, ipconfig /all shows Dynamic Host Configuration Protocol (DHCP) and lease information; ipconfig /release and ipconfig /renew return and request configuration.
- Pool exhaustion, unavailable service, conflicting options, and invalid pool boundaries are different faults. An unapproved Dynamic Host Configuration Protocol (DHCP) server must not be attached to the production Local Area Network (LAN).

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *The router has a DHCP pool of only twenty addresses and several existing leases.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- An authorized router pool contains usable addresses inside its LAN subnet, excluding the router and other managed infrastructure.
- A client set to automatic IP/DNS obtains address, mask, gateway, DNS, and a lease lifetime from the reachable server.
- Discover locates a server, Offer proposes configuration, Request selects it, and ACK commits the lease.
- On authorized Windows equipment, ipconfig /all shows DHCP and lease information; ipconfig /release and ipconfig /renew return and request configuration.
- Pool exhaustion, unavailable service, conflicting options, and invalid pool boundaries are different faults.
- An unapproved DHCP server must not be attached to the production LAN.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of reading a dhcp lease configuration.

Key Takeaway: Automatic configuration includes more than a host address. Lease lifetime, gateway, resolver, and server information help explain which settings the client received.

Slide 8 of 11 DHCP Allocation and Renewal

SIMULATION

📖 **WHAT TO READ & DISPLAY ON SCREEN**

An authorized configuration server allocates unique available addresses from its managed pool. Confirmed leases remain assigned until release, expiry, or a valid lifecycle change.

Interactive Demonstration: `dhcp` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

The router has a DHCP pool of only twenty addresses and several existing leases.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Modeled protocol • configure a pool, step DORA, allocate clients, release or renew, and advance lease time.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of dhcp allocation and renewal.

Key Takeaway: An authorized configuration server allocates unique available addresses from its managed pool. Confirmed leases remain assigned until release, expiry, or a valid lifecycle change.

Slide 9 of 11 DHCP Reservations and Exclusions

SIMULATION

📖 **WHAT TO READ & DISPLAY ON SCREEN**

Reservations associate selected clients with managed addresses, while exclusions remove addresses from general allocation. Active leases and these rules determine whether an address remains available.

Interactive Demonstration: `reservedPoolDemo` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Step through DORA and manage a real simulated pool with leases, renewal, and exhaustion.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Allocate a bounded pool while reserving or excluding selected addresses.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of dhcp reservations and exclusions.

Key Takeaway: Reservations associate selected clients with managed addresses, while exclusions remove addresses from general allocation. Active leases and these rules determine whether an address remains available.

Slide 10 of 11

Thirty laptops join a lab

APPLICATION

📖 WHAT TO READ & DISPLAY ON SCREEN

A working wireless connection does not guarantee an available address lease. When client demand exceeds the free pool, some clients cannot obtain the intended configuration.

- Some new clients cannot obtain an address even though their Wi-Fi association succeeds.
- Inspect pool capacity and active leases before replacing cables.
- An unauthorized second Dynamic Host Configuration Protocol (DHCP) server can supply conflicting options; keep the lab isolated.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *The router has a DHCP pool of only twenty addresses and several existing leases.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Some new clients cannot obtain an address even though their Wi-Fi association succeeds.
- Inspect pool capacity and active leases before replacing cables.
- An unauthorized second DHCP server can supply conflicting options; keep the lab isolated.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of thirty laptops join a lab.

Key Takeaway: A working wireless connection does not guarantee an available address lease. When client demand exceeds the free pool, some clients cannot obtain the intended configuration.

TOPIC 11 OF 13

Topic 11: Router and Wireless Network Configuration

Configure a teaching router, connect clients, and understand LAN, WAN, DHCP, and wireless security.

Curriculum: Networking 1

Excluded: Title (Slide 1), Objectives (Slide 2), and Summary (Final Slide)

Discussion Units: 10 Slides

Slide 3 of 13

Three gateway interface and translation concepts

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A combined gateway connects local clients to another network through separate interface roles. Address translation can change packet addresses, but it does not replace routing or access-control policy.

- Local Area Network (LAN) interface: faces local clients.
- Wide Area Network (WAN) interface: faces another network, usually the provider. Working Internet service also needs provider connectivity and routes.
- Network Address Translation (NAT): translates addresses; a gateway commonly uses port translation so private clients can share a public Internet Protocol version 4 (IPv4) address.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Staff PCs and visitor phones share one provider connection, but visitors should not browse staff folders.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- LAN interface: faces local clients.
- WAN interface: faces another network, usually the provider.
- Working Internet service also needs provider connectivity and routes.
- NAT: translates addresses; a gateway commonly uses port translation so private clients can share a public IPv4 address.
- NAT does not replace firewall policy.
- Local connectivity does not prove WAN or DNS service.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of three gateway interface and translation concepts.

Key Takeaway: A combined gateway connects local clients to another network through separate interface roles. Address translation can change packet addresses, but it does not replace routing or access-control policy.

Slide 4 of 13

SSID, Authentication, and Isolation

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A Service Set Identifier names a wireless network, while authentication controls who can join it. Guest isolation is a separate policy controlling access to private resources.

- Wireless security: avoid open networks, Wired Equivalent Privacy (WEP), and obsolete Wi-Fi Protected Access (WPA)/Temporal Key Integrity Protocol (TKIP). Choose a supported protected configuration.
- Choose a long unique passphrase and protect router administrator credentials separately. Hiding an Service Set Identifier (SSID) is not a security control.
- A guest network should be intentionally isolated from private Local Area Network (LAN) resources. Check the real product's isolation behavior.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Staff PCs and visitor phones share one provider connection, but visitors should not browse staff folders.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Wireless security: avoid open networks, WEP, and obsolete WPA/TKIP.
- Choose a supported protected configuration.
- Choose a long unique passphrase and protect router administrator credentials separately.
- Hiding an SSID is not a security control.
- A guest network should be intentionally isolated from private LAN resources.
- Check the real product's isolation behavior.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of ssid, authentication, and isolation.

Key Takeaway: A Service Set Identifier names a wireless network, while authentication controls who can join it. Guest isolation is a separate policy controlling access to private resources.

Slide 5 of 13

Two protected Wi-Fi security options

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Protected wireless access needs a security mode supported by both access point and clients. Modern personal-network options use suitable authentication and encryption rather than relying on a hidden network name.

- WPA3-Personal: use when supported by the router and clients.
- WPA2-Personal with Advanced Encryption Standard (AES): use for compatible devices that need this protected configuration.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Staff PCs and visitor phones share one provider connection, but visitors should not browse staff folders.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- WPA3-Personal: use when supported by the router and clients.
- WPA2-Personal with AES: use for compatible devices that need this protected configuration.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of two protected wi-fi security options.

Key Takeaway: Protected wireless access needs a security mode supported by both access point and clients. Modern personal-network options use suitable authentication and encryption rather than relying on a hidden network name.

Slide 6 of 13

Example: a laptop joins classroom Wi-Fi

EXAMPLE

📖 WHAT TO READ & DISPLAY ON SCREEN

A laptop needs successful wireless authentication and association before useful network access. Address configuration and upstream service must then work independently of the radio connection.

- The laptop selects Classroom and authenticates using the configured security method. It then obtains a matching address configuration.
- Local services may work while the provider connection is down. Guest isolation can block access to staff resources even when Internet access works.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Configure a teaching router, connect clients, and understand LAN, WAN, DHCP, and wireless security.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- The laptop selects Classroom and authenticates using the configured security method.
- It then obtains a matching address configuration.
- Local services may work while the provider connection is down.
- Guest isolation can block access to staff resources even when Internet access works.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of example: a laptop joins classroom wi-fi.

Key Takeaway: A laptop needs successful wireless authentication and association before useful network access. Address configuration and upstream service must then work independently of the radio connection.

Slide 7 of 13

Wi-Fi Frequencies and Standards

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Wireless networks operate over unlicensed radio frequencies using standardized IEEE 802.11 protocols that balance speed and range.

- 2.4 GHz Frequency Band: Travels farther and penetrates walls well, but offers lower speeds and suffers high congestion from Bluetooth and microwaves.
- 5 GHz Frequency Band: Delivers much faster data speeds and less interference, but has a shorter range and struggles to pass through solid walls.
- Wi-Fi 4 (802.11n): Introduced dual-band operation and MIMO antenna streams, reaching up to 600 Mbps.
- Wi-Fi 5 (802.11ac): High-speed 5 GHz standard providing gigabit wireless speeds for modern laptops and mobile devices.
- Wi-Fi 6 (802.11ax): Modern standard operating on 2.4 GHz, 5 GHz, and 6 GHz (Wi-Fi 6E), optimized for crowded environments like schools and stadiums.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Staff PCs and visitor phones share one provider connection, but visitors should not browse staff folders.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- 2.4 GHz Frequency Band: Travels farther and penetrates walls well, but offers lower speeds and suffers high congestion from Bluetooth and microwaves.
- 5 GHz Frequency Band: Delivers much faster data speeds and less interference, but has a shorter range and struggles to pass through solid walls.
- Wi-Fi 4 (802.11n): Introduced dual-band operation and MIMO antenna streams, reaching up to 600 Mbps.
- Wi-Fi 5 (802.11ac): High-speed 5 GHz standard providing gigabit wireless speeds for modern laptops and mobile devices.
- Wi-Fi 6 (802.11ax): Modern standard operating on 2.4 GHz, 5 GHz, and 6 GHz (Wi-Fi 6E), optimized for crowded environments like schools and stadiums.
- Dual-band home and school routers broadcast on both 2.4 GHz and 5 GHz simultaneously under the same SSID so devices can choose the best connection.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of wi-fi frequencies and standards.

Key Takeaway: Wireless networks operate over unlicensed radio frequencies using standardized IEEE 802.11 protocols that balance speed and range.

Slide 8 of 13

Coordinating Router, Pool, and Wi-Fi

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Router settings affect both management access and client connectivity. A compatible local address, allocation pool, and wireless configuration reduce avoidable disconnections.

- Choose a private Local Area Network (LAN) subnet, then a pool of usable addresses inside it excluding the router's address.
- Changing the router Local Area Network (LAN) address can interrupt management access; reconnect and obtain a lease in the new subnet.
- Save settings, connect one wired and one wireless client, then verify Internet Protocol (IP) options, gateway reachability, and service access separately.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Staff PCs and visitor phones share one provider connection, but visitors should not browse staff folders.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Choose a private LAN subnet, then a pool of usable addresses inside it excluding the router's address.
- Changing the router LAN address can interrupt management access; reconnect and obtain a lease in the new subnet.
- Save settings, connect one wired and one wireless client, then verify IP options, gateway reachability, and service access separately.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of coordinating router, pool, and wi-fi.

Key Takeaway: Router settings affect both management access and client connectivity. A compatible local address, allocation pool, and wireless configuration reduce avoidable disconnections.

Slide 9 of 13

Applying and Testing Router Settings

WALKTHROUGH

📖 WHAT TO READ & DISPLAY ON SCREEN

A router change must be followed by reconnecting clients and checking their received settings. Local access, name resolution, and upstream service provide different verification evidence.

- Use an isolated, authorized router and record its current settings. A wired management connection reduces dependence on changing Wi-Fi settings.
- Choose a private Local Area Network (LAN) subnet and a matching pool excluding the router address. A LAN address change can interrupt management and existing leases.
- Set an Service Set Identifier (SSID) and supported WPA3-Personal or WPA2-AES security. Use a long unique passphrase distinct from router administrator credentials.
- Reconnect a wired and an authenticated wireless client, then inspect assigned address, mask, gateway, and Domain Name System (DNS).
- Check local access before Wide Area Network (WAN) services. Guest isolation depends on the actual device features and policy; restore approved settings after a physical demonstration.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Staff PCs and visitor phones share one provider connection, but visitors should not browse staff folders.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Use an isolated, authorized router and record its current settings.
- A wired management connection reduces dependence on changing Wi-Fi settings.
- Choose a private LAN subnet and a matching pool excluding the router address.
- A LAN address change can interrupt management and existing leases.
- Set an SSID and supported WPA3-Personal or WPA2-AES security.
- Use a long unique passphrase distinct from router administrator credentials.
- Reconnect a wired and an authenticated wireless client, then inspect assigned address, mask, gateway, and DNS.
- Check local access before WAN services.
- Guest isolation depends on the actual device features and policy; restore approved settings after a physical demonstration.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of applying and testing router settings.

Key Takeaway: A router change must be followed by reconnecting clients and checking their received settings. Local access, name resolution, and upstream service provide different verification evidence.

Slide 10 of 13 Router LAN and Wi-Fi Workbench

SIMULATION

📖 **WHAT TO READ & DISPLAY ON SCREEN**

A small gateway can combine local addressing, automatic configuration, and wireless access. Those settings must agree before clients can use the intended services.

Interactive Demonstration: `router` — Use on-screen controls to step through or demonstrate the concept visually.

💡 **BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)**

🌟 **PRACTICAL CONTEXT / INTUITION**

Staff PCs and visitor phones share one provider connection, but visitors should not browse staff folders.

🗣️ **HOW TO EXPLAIN IT (TALKING POINTS)**

- Browser-only model • configure LAN and Wi-Fi, apply settings, then add wired or authenticated wireless clients.

❓ **QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)**

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of router lan and wi-fi workbench.

Key Takeaway: A small gateway can combine local addressing, automatic configuration, and wireless access. Those settings must agree before clients can use the intended services.

Slide 11 of 13 Guest Wi-Fi Isolation and Internet

SIMULATION

📖 **WHAT TO READ & DISPLAY ON SCREEN**

Guest isolation can permit Internet use while blocking access to staff resources. Wireless association, isolation policy, local services, and the upstream connection remain separate conditions.

Interactive Demonstration: `guestIsolationDemo` — Use on-screen controls to step through or demonstrate the concept visually.

💡 **BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)**

🌟 **PRACTICAL CONTEXT / INTUITION**

Configure a teaching router, connect clients, and understand LAN, WAN, DHCP, and wireless security.

🗣️ **HOW TO EXPLAIN IT (TALKING POINTS)**

- Change guest association, isolation policy, WAN state, and destination to separate connectivity conditions.

❓ **QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)**

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of guest wi-fi isolation and internet.

Key Takeaway: Guest isolation can permit Internet use while blocking access to staff resources. Wireless association, isolation policy, local services, and the upstream connection remain separate conditions.

Slide 12 of 13

A small training center

APPLICATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Staff and visitors may share an Internet connection without sharing access to internal files. A correctly configured guest network separates those access needs.

- Use a private staff Local Area Network (LAN) and a properly isolated guest Wireless Local Area Network (WLAN) supported by the equipment.
- Document Wide Area Network (WAN) requirements from the provider instead of guessing settings.
- Confirm guests cannot access staff resources while approved Internet access remains possible.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Staff PCs and visitor phones share one provider connection, but visitors should not browse staff folders.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Use a private staff LAN and a properly isolated guest WLAN supported by the equipment.
- Document WAN requirements from the provider instead of guessing settings.
- Confirm guests cannot access staff resources while approved Internet access remains possible.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of a small training center.

Key Takeaway: Staff and visitors may share an Internet connection without sharing access to internal files. A correctly configured guest network separates those access needs.

TOPIC 12 OF 13

Topic 12: Network File and Printer Sharing

Publish a modeled resource and see how share permissions and file-system permissions combine.

Curriculum: Networking 1 **Excluded:** Title (Slide 1), Objectives (Slide 2), and Summary (Final Slide) **Discussion Units:** 8 Slides

Slide 3 of 11

What Is a Network Share?

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A network share makes a resource available to authorized clients through a hosting device. The host and required service must remain reachable for remote access.

- In peer-to-peer sharing, a normal Personal Computer (PC) may host the resource; it must stay on and reachable.
- A dedicated file/print server centralizes resource management and availability.
- Windows discovery helps find hosts but direct paths such as \\LAB-PC\ClassFiles may work even when browsing does not. Domain Name System (DNS) or other name resolution must resolve the host name.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *An organization distributes read-only reference files and grants editing only to named contributors.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- In peer-to-peer sharing, a normal PC may host the resource; it must stay on and reachable.
- A dedicated file/print server centralizes resource management and availability.
- Windows discovery helps find hosts but direct paths such as \\LAB-PC\ClassFiles may work even when browsing does not.
- DNS or other name resolution must resolve the host name.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of what is a network share?.

Key Takeaway: A network share makes a resource available to authorized clients through a hosting device. The host and required service must remain reachable for remote access.

Slide 4 of 11

Two permission gates for remote file access

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Remote access to a Windows file share depends on share permissions and underlying file-system permissions. Both gates must allow the requested operation for access to succeed.

- Share permissions: control the access allowed through the published network share.
- File-system permissions: New Technology File System (NTFS) permissions control access to the underlying files and folders.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *An organization distributes read-only reference files and grants editing only to named contributors.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Share permissions: control the access allowed through the published network share.
- File-system permissions: NTFS permissions control access to the underlying files and folders.
- Remote access is limited by both sets.
- Read/viewer permits reading; change/editor permits modification only when NTFS also allows it.
- Actual ACLs also include groups, inheritance, and explicit denies.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of two permission gates for remote file access.

Key Takeaway: Remote access to a Windows file share depends on share permissions and underlying file-system permissions. Both gates must allow the requested operation for access to succeed.

Slide 5 of 11

Example: students read a shared reference folder

EXAMPLE

📖 WHAT TO READ & DISPLAY ON SCREEN

A shared folder can permit reading while denying remote edits. The effective result depends on both permission sets rather than the more generous setting alone.

- Read at the share and Modify at the file system permit reading but do not permit remote edits in this simplified example.
- Effective access is limited by both permission gates. A reachable host does not itself grant access to its files.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Publish a modeled resource and see how share permissions and file-system permissions combine.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Read at the share and Modify at the file system permit reading but do not permit remote edits in this simplified example.
- Effective access is limited by both permission gates.
- A reachable host does not itself grant access to its files.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of example: students read a shared reference folder.

Key Takeaway: A shared folder can permit reading while denying remote edits. The effective result depends on both permission sets rather than the more generous setting alone.

Slide 6 of 11

Securing Shared Files and Printers

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Sharing should give named users only the access their work requires. A reachable host still needs suitable identity, permission, service, and client support.

- Create a test folder → Properties → Sharing → Advanced Sharing; share only the intended folder with named authorized users.
- Check Security permissions separately. Enable discovery/file sharing only for the approved profile and retain appropriate firewall rules.
- Install a supported printer driver and authorize the printer share. A reachable Personal Computer (PC) does not guarantee spooler, driver, or permission success.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *An organization distributes read-only reference files and grants editing only to named contributors.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Create a test folder → Properties → Sharing → Advanced Sharing; share only the intended folder with named authorized users.
- Check Security permissions separately.
- Enable discovery/file sharing only for the approved profile and retain appropriate firewall rules.
- Install a supported printer driver and authorize the printer share.
- A reachable PC does not guarantee spooler, driver, or permission success.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of securing shared files and printers.

Key Takeaway: Sharing should give named users only the access their work requires. A reachable host still needs suitable identity, permission, service, and client support.

Slide 7 of 11

Publishing and Accessing a Windows Shared Folder

WALKTHROUGH

📖 WHAT TO READ & DISPLAY ON SCREEN

Publishing a folder creates a network access path to a resource on the host. Remote verification must check the intended client identity and both permission layers.

- On a trusted approved network, a host shares a non-sensitive folder through Properties → Sharing → Advanced Sharing with a named authorized identity.
- The Security tab controls New Technology File System (NTFS) permissions separately. For remote file access, both share and file-system rules must allow the requested action.
- From another computer, use a Universal Naming Convention (UNC) path such as \\HOST\ClassFiles and authenticate with the intended account. Host-side access does not verify the client identity.
- Read-only access permits reading without edits; editing requires suitable permissions at both layers. Retain approved discovery/file-sharing firewall rules.
- Remove temporary shares after a physical demonstration. Printer sharing additionally depends on print permission, driver compatibility, and a working host/spooler.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *An organization distributes read-only reference files and grants editing only to named contributors.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- On a trusted approved network, a host shares a non-sensitive folder through Properties → Sharing → Advanced Sharing with a named authorized identity.
- The Security tab controls NTFS permissions separately.
- For remote file access, both share and file-system rules must allow the requested action.
- From another computer, use a UNC path such as \\HOST\ClassFiles and authenticate with the intended account.
- Host-side access does not verify the client identity.
- Read-only access permits reading without edits; editing requires suitable permissions at both layers.
- Retain approved discovery/file-sharing firewall rules.
- Remove temporary shares after a physical demonstration.
- Printer sharing additionally depends on print permission, driver compatibility, and a working host/spooler.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of publishing and accessing a windows shared folder.

Key Takeaway: Publishing a folder creates a network access path to a resource on the host. Remote verification must check the intended client identity and both permission layers.

Slide 8 of 11 Share vs NTFS Permissions

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Effective remote permissions are limited by every applicable access gate in this simplified model. Network reachability alone cannot override an unshared resource or denied operation.

Interactive Demonstration: `sharing` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

An organization distributes read-only reference files and grants editing only to named contributors.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Modeled permissions • select host and client, publish a folder, choose both permission levels, and attempt access.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of share vs ntfs permissions.

Key Takeaway: Effective remote permissions are limited by every applicable access gate in this simplified model. Network reachability alone cannot override an unshared resource or denied operation.

Slide 9 of 11 How Shared Printer Queues Work

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

A shared printer queue retains accepted jobs and processes their pages in arrival order. Device outages pause processing, while submission permissions determine whether new jobs can enter the queue.

Interactive Demonstration: `printQueueDemo` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

Publish a modeled resource and see how share permissions and file-system permissions combine.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Submit modeled jobs and observe FIFO page processing, retained queues, and new-job permissions.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of how shared printer queues work.

Key Takeaway: A shared printer queue retains accepted jobs and processes their pages in arrival order. Device outages pause processing, while submission permissions determine whether new jobs can enter the queue.

Slide 10 of 11

Read-Only vs Editable Sharing

APPLICATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Reference files and editable working documents can require different access policies. Named accounts and remote permission checks help keep changes limited to authorized contributors.

- Avoid making one broad writable share for sensitive documents.
- Use named accounts and verify permissions from a remote client.
- A shared printer also needs a reachable host, driver, and permission to print.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

An organization distributes read-only reference files and grants editing only to named contributors.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Avoid making one broad writable share for sensitive documents.
- Use named accounts and verify permissions from a remote client.
- A shared printer also needs a reachable host, driver, and permission to print.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of read-only vs editable sharing.

Key Takeaway: Reference files and editable working documents can require different access policies. Named accounts and remote permission checks help keep changes limited to authorized contributors.

TOPIC 13 OF 13

Topic 13: Network Testing and Troubleshooting

Inspect evidence, run state-derived diagnostic commands, and repair nine common network faults.

Curriculum: Networking 1 **Excluded:** Title (Slide 1), Objectives (Slide 2), and Summary (Final Slide) **Discussion Units:** 9 Slides

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Systematic Troubleshooting Process

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Troubleshooting starts by defining the symptom and collecting evidence. A justified change is followed by repeating the original failing test and documenting the result.

- Start with power, cable, port, and wireless association. Compare with a known-good link when authorized.
- Check address, mask, duplicate conflicts, Dynamic Host Configuration Protocol (DHCP) service, and gateway. Test local and remote reachability separately.
- If numeric connectivity works but names fail, investigate Domain Name System (DNS). If the host is reachable but a share is denied, inspect identity and permissions.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A student can reach the gateway by IP, but cannot open a website by its name.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Start with power, cable, port, and wireless association.
- Compare with a known-good link when authorized.
- Check address, mask, duplicate conflicts, DHCP service, and gateway.
- Test local and remote reachability separately.
- If numeric connectivity works but names fail, investigate DNS.
- If the host is reachable but a share is denied, inspect identity and permissions.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of systematic troubleshooting process.

Key Takeaway: Troubleshooting starts by defining the symptom and collecting evidence. A justified change is followed by repeating the original failing test and documenting the result.

Slide 4 of 12

Six Windows diagnostic commands

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Diagnostic commands inspect configuration, reachability, name resolution, or neighbor information. Each answers a limited question rather than proving that the whole network works.

- ipconfig: shows basic interface configuration; it does not prove reachability.
- ipconfig /all: shows detail such as Domain Name System (DNS), Dynamic Host Configuration Protocol (DHCP) server, and leases; it does not prove reachability.
- ping: tests Internet Control Message Protocol (ICMP) Echo behavior and round-trip time; a timeout can also mean filtering or packet loss.
- tracert: provides TTL-based hop evidence; filtered probes can produce gaps.
- nslookup: queries Domain Name System (DNS); success does not prove the destination service works.
- arp -a: lists cached Internet Protocol version 4 (IPv4) neighbor mappings, not every device on the Local Area Network (LAN).

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A student can reach the gateway by IP, but cannot open a website by its name.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- ipconfig: shows basic interface configuration; it does not prove reachability.
- ipconfig /all: shows detail such as DNS, DHCP server, and leases; it does not prove reachability.
- ping: tests ICMP Echo behavior and round-trip time; a timeout can also mean filtering or packet loss.
- tracert: provides TTL-based hop evidence; filtered probes can produce gaps.
- nslookup: queries DNS; success does not prove the destination service works.
- arp -a: lists cached IPv4 neighbor mappings, not every device on the LAN.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of six windows diagnostic commands.

Key Takeaway: Diagnostic commands inspect configuration, reachability, name resolution, or neighbor information. Each answers a limited question rather than proving that the whole network works.

Slide 5 of 12

Nine modeled network faults

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Similar symptoms can come from different physical, addressing, service, or permission faults. Distinguishing the cause requires evidence tied to the failed operation.

- Ethernet cable disconnected.
- Incorrect static Internet Protocol (IP) address.
- Incorrect subnet mask.
- Duplicate Internet Protocol (IP) address.
- Dynamic Host Configuration Protocol (DHCP) server unavailable.
- Incorrect default gateway.
- Domain Name System (DNS) server problem.
- File-sharing permission denied.
- Wireless client disconnected.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A student can reach the gateway by IP, but cannot open a website by its name.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Ethernet cable disconnected.
- Incorrect static IP address.
- Incorrect subnet mask.
- Duplicate IP address.
- DHCP server unavailable.
- Incorrect default gateway.
- DNS server problem.
- File-sharing permission denied.
- Wireless client disconnected.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of nine modeled network faults.

Key Takeaway: Similar symptoms can come from different physical, addressing, service, or permission faults. Distinguishing the cause requires evidence tied to the failed operation.

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Example: numeric access works but a name fails

EXAMPLE

📖 WHAT TO READ & DISPLAY ON SCREEN

A successful numeric service test and a failed name query point to different functional results. Name-resolution settings and reachability deserve investigation before replacing an already working link.

- If the same service works using its numeric address while name resolution times out, investigate resolver settings and name-service reachability.
- Do not replace a working cable based on this symptom alone. Correct the identified resolver fault, then repeat the original name-based request.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Inspect evidence, run state-derived diagnostic commands, and repair nine common network faults.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- If the same service works using its numeric address while name resolution times out, investigate resolver settings and name-service reachability.
- Do not replace a working cable based on this symptom alone.
- Correct the identified resolver fault, then repeat the original name-based request.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of example: numeric access works but a name fails.

Key Takeaway: A successful numeric service test and a failed name query point to different functional results. Name-resolution settings and reachability deserve investigation before replacing an already working link.

Slide 7 of 12

Understanding Traceroute Results

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Traceroute uses limited-lifetime probes to obtain evidence from successive hops. A missing reply can reflect filtering, while an explicit unreachable response provides different evidence about the path.

Interactive Demonstration: `ttlTraceDemo` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *Inspect evidence, run state-derived diagnostic commands, and repair nine common network faults.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Inspect modeled ICMP TTL-expiry evidence, hidden replies, and an explicit broken-link report.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of understanding traceroute results.

Key Takeaway: Traceroute uses limited-lifetime probes to obtain evidence from successive hops. A missing reply can reflect filtering, while an explicit unreachable response provides different evidence about the path.

Slide 8 of 12

Verifying a Network Repair

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A troubleshooting hypothesis should explain the available evidence and predict a useful test. A repair is confirmed by repeating the original failed operation, not by assuming that a changed setting solved it.

- 169.254.x.x suggests link-local configuration; check Dynamic Host Configuration Protocol (DHCP) reachability rather than guessing a public address.
- A wrong gateway can break remote access while local peers still work. A Domain Name System (DNS) fault can leave numeric connectivity intact.
- Document the original state, corrective action, and successful retest. Escalate unfamiliar equipment faults instead of making uncontrolled changes.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A student can reach the gateway by IP, but cannot open a website by its name.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- 169.254.x.x suggests link-local configuration; check DHCP reachability rather than guessing a public address.
- A wrong gateway can break remote access while local peers still work.
- A DNS fault can leave numeric connectivity intact.
- Document the original state, corrective action, and successful retest.
- Escalate unfamiliar equipment faults instead of making uncontrolled changes.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of verifying a network repair.

Key Takeaway: A troubleshooting hypothesis should explain the available evidence and predict a useful test. A repair is confirmed by repeating the original failed operation, not by assuming that a changed setting solved it.

Slide 9 of 12

Troubleshooting Workflow: Symptom to Fix

WALKTHROUGH

📖 WHAT TO READ & DISPLAY ON SCREEN

A complete investigation connects the reported symptom to observations, a likely cause, and a targeted correction. The final retest checks whether the user's original operation now succeeds.

- Define the symptom and scope, then inspect power, physical link, association, and the known healthy configuration.
- ipconfig /all exposes address, mask, gateway, Domain Name System (DNS), and Dynamic Host Configuration Protocol (DHCP) details without proving reachability.
- Compare a local numeric test, a gateway test, a remote numeric test, and a Domain Name System (DNS) query to narrow the affected function.
- Use tracert for hop evidence and arp -a for cached Internet Protocol version 4 (IPv4) neighbors, recognizing filtering and cache limitations.
- Change one justified setting, repeat the original failing test, and document the result. Keep security controls active and escalate unfamiliar equipment faults.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A student can reach the gateway by IP, but cannot open a website by its name.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Define the symptom and scope, then inspect power, physical link, association, and the known healthy configuration.
- ipconfig /all exposes address, mask, gateway, DNS, and DHCP details without proving reachability.
- Compare a local numeric test, a gateway test, a remote numeric test, and a DNS query to narrow the affected function.
- Use tracert for hop evidence and arp -a for cached IPv4 neighbors, recognizing filtering and cache limitations.
- Change one justified setting, repeat the original failing test, and document the result.
- Keep security controls active and escalate unfamiliar equipment faults.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of troubleshooting workflow: symptom to fix.

Key Takeaway: A complete investigation connects the reported symptom to observations, a likely cause, and a targeted correction. The final retest checks whether the user's original operation now succeeds.

Slide 10 of 12

Diagnosing Network Faults Workbench

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Different command results expose different parts of the network state. Comparing them supports a targeted repair rather than unrelated configuration changes.

Interactive Demonstration: `troubleshoot` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A student can reach the gateway by IP, but cannot open a website by its name.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Modeled network • nine deterministic fault states.
- Inspect commands, select a cause, apply a targeted repair, and retest.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of diagnosing network faults workbench.

Key Takeaway: Different command results expose different parts of the network state. Comparing them supports a targeted repair rather than unrelated configuration changes.

Slide 11 of 12

Local LAN vs Internet vs DNS

APPLICATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Reaching a local gateway does not prove that remote routes, name resolution, or a website work. Separate tests identify which stage succeeds and which still needs investigation.

- Check numeric external reachability and Domain Name System (DNS) responses to separate routing from name resolution.
- Do not assume the cable is faulty when the gateway is responding.
- Even working Domain Name System (DNS) and ping do not prove a web service or permission is functioning.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 PRACTICAL CONTEXT / INTUITION

> *A student can reach the gateway by IP, but cannot open a website by its name.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Check numeric external reachability and DNS responses to separate routing from name resolution.
- Do not assume the cable is faulty when the gateway is responding.
- Even working DNS and ping do not prove a web service or permission is functioning.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "How does this concept apply when configuring or troubleshooting a real network?"

Expected Answer / Guidance: Encourages learners to explain the functional purpose and practical impact of local lan vs internet vs dns.

Key Takeaway: Reaching a local gateway does not prove that remote routes, name resolution, or a website work. Separate tests identify which stage succeeds and which still needs investigation.